EXPO CITY DUBAI

Expo Hills Phase 1

Developed Design

Fire and Life Safety Report

C3085-RPT-3EH6105-FP-0000001(5)

25 Sep 2026

KEO

SHE



EXPO CITY DUBAI

FIRE AND LIFE SAFETY

100% DEVELOPED DESIGN REPORT

KEO

SHE

TABLE OF CONTENTS

01. INTRODUCTION
02. CONSTRUCTION AND COMPARTMENTATION
03. MEANS OF EGRESS
04. BALCONY TERRACE AND WINDOW
05. DOOR
06. OCCUPANT LOAD/ EXIT WIDTH
07. TRAVEL DISTANCE
08. FIRE FIGHTING ACCESS
09. FIRE FIGHTING SYSTEMS
10. SMOKE MANAGEMENT SYSTEMS
11. LPG SYSTEMS
12. FIRE DETECTION & ALARM SYSTEM
13. EMERGENCY LIGHTING & EXIT SIGNAGE
14. STANDBY EMERGENCY POWER
15. SOLAR POWER GENERATION SYSTEMS
16. INFORMATION USED & LIMITATION OF THIS REPORT

INTRODUCTION

This report is intended for Expo Hill Residential Development located at Expo Hills, Dubai, UAE. The report should be read in conjunction with the FLS review comments shared during this stage. The proposed development will consist of 8 Buildings (A1-A8), all buildings will be sharing common basement, ground and first level and above 8 to 12 typical levels, refer to Render figure below:



Figure 1: Overall Render

Objectives of Report

This report will address the fire and life safety requirements which are required by the Authority Having Jurisdiction (AHJ) and the applicable codes. The report will assess the requirements based on UAE Fire Code 2018 editions and NFPA standards. This report is not intended for submission to the authorities.

Authority Having Jurisdiction

The Authority Having Jurisdiction (AHJ) is an organization, office, or individual responsible for enforcing the requirements of a code or standard, or for approving equipment, materials, and installation or procedure. Dubai Civil Defence is the AHJ for the Fire & Life Safety Design of this project.

Applicable Codes, Standards and Regulations

The project will be assessed based on the local authority requirements and the applicable building safety codes and standards as listed below. Dubai Civil Defence (DCD) is the Authority Having Jurisdiction (AHJ) for this project. The primary life safety code is the UAE Fire and Life Safety Code of Practice (UAEFC-2018) and it is our understanding from previous experience that DCD permits the use of international standards such as NFPA 5000 and NFPA 101 as supplementary design guidelines. Where there are inconsistencies between UAEFC and NFPA, the local requirements will be applicable for this project.

General Project Characteristics

Buildings Characteristics - Tallest Building	
Number of Storeys	13 (including Ground)
Building Height	48.9m >23m (roof top)
Classification of Building	High-rise

Table 1: Buildings Characteristics

CONSTRUCTION AND COMPARTMENTATION

Occupancy Classification	Horizontal Separation in m				Permitted % of Opening Protectives
	0 to 1.5 m	1.5 to 3 m	3 to 9 m	> 9 m	
Residential, Assembly,	1	1	0	0	Refer to Section 2.8.5 of the UAE code, if required

Table 2: Fire Resistance Rating of Exterior walls

Fire Rating Requirements with respect to Property lines or other Buildings

Based on NFPA 5000, and Section 2.7 of the 2018 UAE Fire and Life Safety Code, buildings will be required to maintain a minimum horizontal distance from each other. If the minimum distance is not met, any exterior wall within this distance will be required to be fire-rated and unprotected openings in that building face will be required to be protected or limited in size. Where two or more buildings are located on the same plot, the horizontal separation will be measured from the building face to an imaginary line drawn between the exterior walls of the adjacent buildings. The relevant parts of Table 1.3 of 2018 UAE Fire and Life Safety Code are reproduced in the following table:

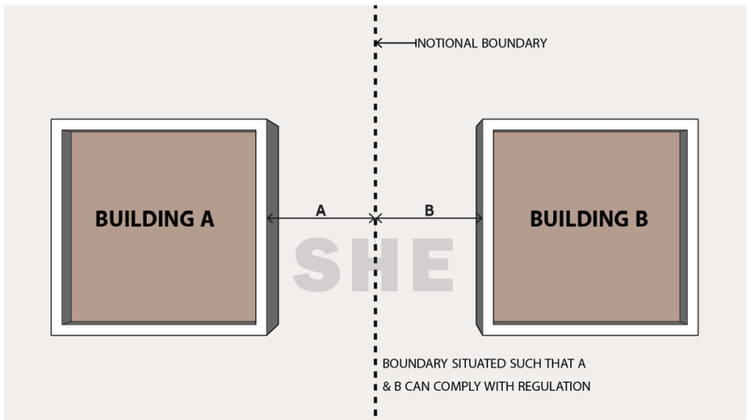


Figure 2: Measurement of Horizontal Distance

Structural Requirements

Construction type is determined in accordance with NFPA 5000 requirements and UAE Fire Code, Chapter 1, Table 1.7. Considering the area of the largest floor, number of storey and height of each building, applicable construction type shall be Type I (332).

The construction type for the proposed building shall be as given in Table below.

Description	Building (tallest building)
Building Height (m)	48.9m >23 m (From road level to the top of the roof)
Number of Storeys	13 (including Ground)
UAEFC and NFPA	TYPE I (332)

Table 3: Type of Construction Requirements

Type I construction will consist of construction in which the structural members, including walls, columns, beams, girders, trusses, arches, floors, and roofs, are of approved non-combustible materials. The structural elements of the building must achieve the minimum level of fire rating indicated in below table. Stairs should be constructed of RCC (reinforced concrete).

Structural Elements		Fire Rating
Exterior bearing walls	Supporting more than one floor	3 (Hours)
	Supporting one floor only	3 (Hours)
	Supporting a roof only	3 (Hours)
Interior bearing walls	Supporting more than one floor	3 (Hours)
	Supporting one floor only	2 (Hours)
	Supporting a roof only	2 (Hours)
Columns	Supporting more than one floor	3 (Hours)
	Supporting one floor only	2 (Hours)
	Supporting a roof only	2 (Hours)
Beams,Girders, Trusses and Arches	Supporting more than one floor	3 (Hours)
	Supporting one floor only	2 (Hours)
	Supporting a roof only	2 (Hours)
Floor-Ceiling Assemblies		2 (Hours)
Roof-Ceiling Assemblies		1.5 (Hours)
Interior Non-Bearing Walls		0 (Hours)
Exterior Non-Bearing Walls		0 (Hours)

Table 4: Fire-Ratings of Structural Elements - TYPE I (332)

It is assumed that building core structural elements are constructed of concrete and there are no exposed steel structural elements. Any exposed steel structural elements must be provided with passive fire protection elements. For concrete construction, the appropriate concrete cover will be required to be provided.

Other Fire Separations

There are required fire separations other than the structural elements fire rating for the assessed building. These fire separations shall be provided for shafts, occupancy group separations, public multi-tenant corridors and stores as indicated in the Table below. Any openings such as doors and windows located inside fire-rated construction must also be provided with the required fire-resistance rating.

It is noted that in case a compartment fire rating and structural fire rating applies to the same fire separation, the higher rating will be applicable.

The Table below summarizes the fire-resistance ratings of walls and fire rated doors to be provided for other fire separations of the building in accordance with Chapter 1, Tables 1.9, 1.10, 1.11.a, 1.11b of UAE Fire Code.

Fire Resistance Separation (with Automatic Fixed Fire Protection System)		
Components	**Walls and partitions**	**Fire Door Assembly**
Vertical Shafts	2-hours	90-Minutes
Exit Stair Enclosures, Elevator Shafts	2-hours	90-Minutes
Exit Access Corridor (residential)	1-Hour	60-Minutes
Between dwelling units	1-Hour	60-Minutes
Between Car Park and Adjacent Occupancies	2-Hours	90-Minutes
Generator Room	2-Hours	90-Minutes
Fire Command Centre (FCC) Room	1-Hour	60-Minutes
Store Rooms > 9.3 m^2	1-Hour	60-Minutes
L.V. Room (Protected with applicable fire protection systems)	2-Hours	90-Minutes
ETS Room	0-Hour	0-Minutes
Fire Pump Room	2-Hours	90-Minutes
Telephone Room	1-Hour	60-Minutes
Lift M/C Room	2-Hours	90-Minutes
Garbage Room (Typical Floor)	1-Hour	60-Minutes
Garbage Room with Chute (Ground Floor)	2-Hours	90-Minutes
Electrical Room	1-Hour	60-Minutes
Substation Room	2-Hours	90-Minutes

Table 5: Fire Resistance Separation

It is noted that in case a compartment fire rating and structural fire rating applies to the same fire wall or compartment, the higher rating should be applicable. For examples, for any stair enclosure as part of the core support, the higher fire rating for the structure should be applied as per type of construction assessment. However, the stair enclosure opening is only required to be maintained for 90 minutes.

Fire Protection Rating of Windows and Doors

Every opening in a fire separation/ fire barrier should be protected to limit the spread of fire and to restrict the movement of smoke from one side of the fire barrier to the other. Fire doors in all fire separations should be provided with fire-protection ratings in accordance with UAE Fire Code, Chapter 1, Table 1.10. Refer to Table 7 of this report for more details.

Fire windows If provided, the combined area of glazing in fire rated window assemblies and fire rated door assemblies used in fire barriers should not exceed 25 percent of the area of the fire barrier that is common with any room.

All rated doors should be provided with self-closing devices and latching. Panic hardware if provided should comply with UAE Fire Code.

Penetrations and Openings into Exit Stair Enclosure

Exit enclosures should not have any openings except for openings required for services, serving the exit facility. Moreover, any openings in the exit enclosures should limit a direct opening into

mechanical plant spaces or storage spaces.

Penetrations into, and openings through, an exit enclosure assembly should be limited to the following:

- Doors from corridors and door assemblies for egress from the enclosure
- Required exit doors
- Fire Hose & Sprinkler piping
- Fire Protection piping valves
- Fire detection and alarm system wiring enclosed in metal conduits.
- Electrical conduits serving the stairway

Fire Dampers

In accordance with the UAE Fire Code, Fire and/or smoke dampers should be provided where ductwork penetrates or terminates at openings in walls or partitions required to achieve a fire or smoke resistance performance. All fire and/or smoke dampers should have a fire rating that is equivalent to the performance rating of the wall or partition in which it is to be installed.

Where a fire barrier is also constructed as a smoke partition, a combination fire/smoke damper designed and tested in accordance with the requirements of ANSI/UL 555, Standard for Fire Dampers, and ANSI/UL 555S should be installed.

Motorised fire / smoke rated dampers should be provided within the air supply system to separate the return air portion in the event of a fire.

The minimum fire protection rating, as specified in Table 1.13 of the UAE Fire Code and reproduced in table below.

Fire Damper Rating	
Fire Resistance Rating of Assembly	**Minimum Damper Rating**
3-hours or greater fire resistance–rated assemblies	3-Hours
Less than 3-hours fire resistance–rated assemblies	90-minutes

Table 6: Fire Damper Rating

In accordance with Chapter 1, Section 2.15.1.3, UAE Fire Code, fire dampers should be required in the following locations:

- Ducts and air-transfer openings, penetrating walls or partitions having a fire resistance rating of 2 or more hours
- Ducts and air-transfer openings, penetrating shaft walls having a fire resistance rating of 1 or more hours
- Ducts and air-transfer openings, penetrating floors that are required to have protected openings where the duct also is not protected by a shaft enclosure
- Air-transfer openings that occur in walls or partitions that are required to have a fire-resistive rating of 30-minutes or more.

Smoke Dampers

In accordance with Table 10.1, Chapter 10 of UAE Fire Code, the following are the areas where Smoke Dampers are required.

- HVAC duct branching serving apartments, dwelling units.
- Smoke dampers should be installed at or adjacent to the point where air ducts pass through required smoke barrier, but in no case smoke damper shall be installed more than 0.6m from the barrier, or after the first air duct inlet or outlet, whichever is close to the barrier.
- If the air handling equipment is serving more than one floor then smoke damper shall be installed in system with a capacity greater than 7080L/s (15,000ft3/min) to isolate the air handling equipment, including filters from the remainder of the system on both the building supply and return side, in order to restrict the circulation of smoke.
- Smoke damper used for protection of openings in smoke barriers or in engineered smoke control systems shall be classified in accordance with ANSI/UL 555S, and with minimum class II leakage rating, and elevated temperature rating shall not be less than 2500 C

Where smoke barrier is also constructed and serving as Fire barrier/ Wall, a combination (Smoke/Fire) Damper will be required be installed for all Ducts and air-transfer openings penetrating such barriers/Walls.

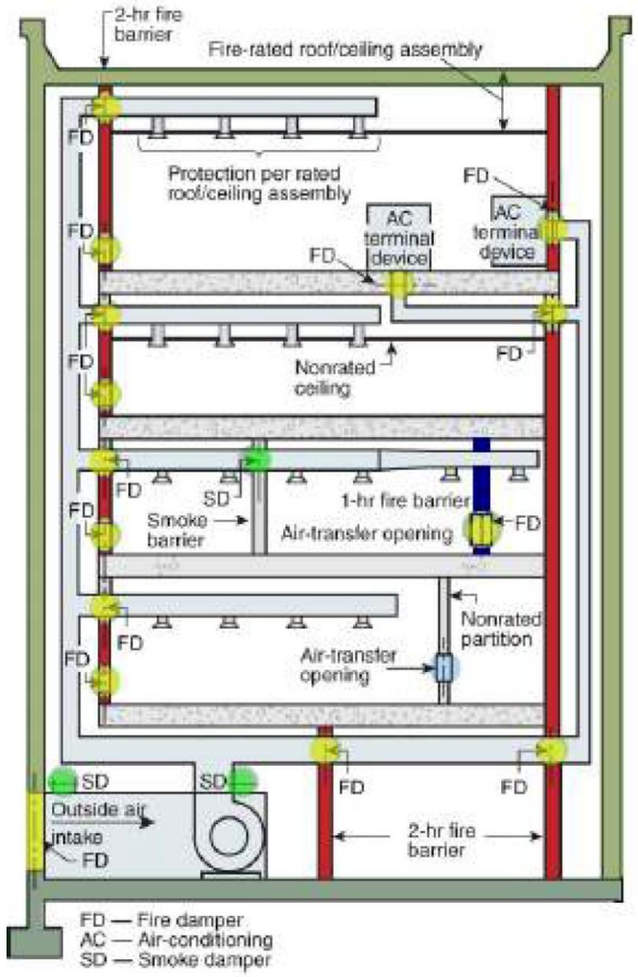


Figure 3: Fire Dampers (FD) and Smoke Damper (SD) in various penetrations

Building Services in Fire Separations

Penetrations for cables, cable trays, conduits, pipes, tubes, combustion vents and exhaust vents, wires, and similar items to accommodate electrical, mechanical, plumbing, and communications systems that pass through a wall, floor, or floor-ceiling assembly constructed as a fire separation, will be required to be protected by a fire-stop system or device. The fire-stop system or device will be tested in accordance with ASTM E 814, "Standard Test Method for Fire Tests of Through-Penetration Fire Stops", or UL 1479, "Standard for Safety for Fire Tests of Through-Penetration Fire Stops". Fire-stop systems will not be required to be provided where concrete, grout, or mortar has been used to fill the annular spaces around cast-iron, copper, or steel piping, or steel conduit or tubing, that penetrates one or more concrete or masonry fire resistance-rated assemblies.

The nominal diameter of each penetrating item will not be permitted to exceed 150 mm, and the opening size will not be permitted to exceed 92,909 mm2 (1 ft2). The thickness of the concrete, grout, or mortar will be required to be the full thickness of the assembly. Examples of fire stop are indicated in the Figures below:

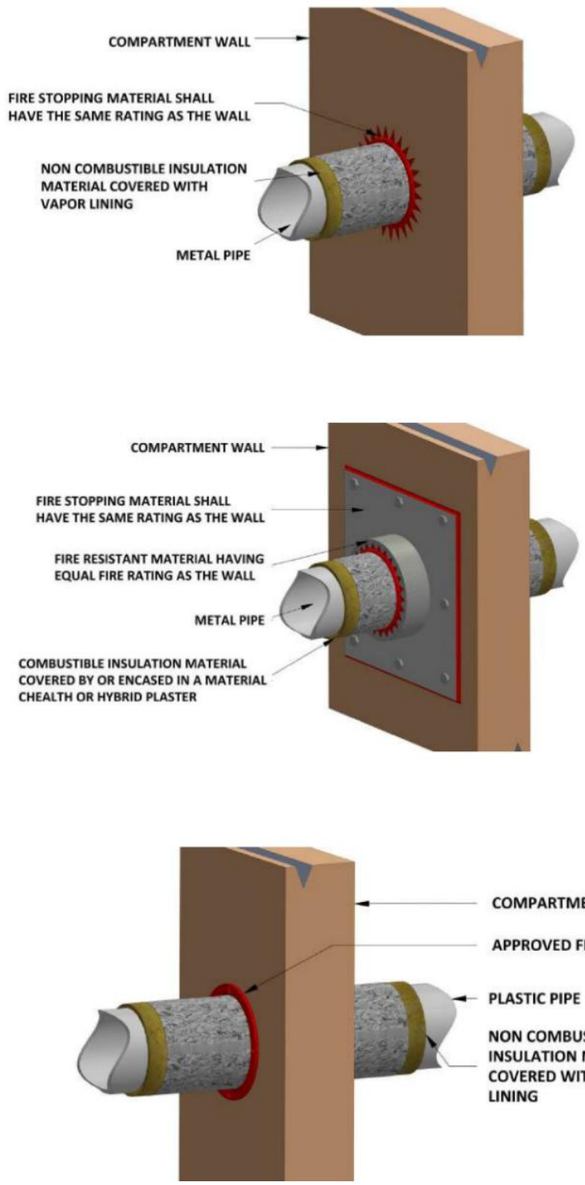


Figure 4: Pipes with combustible Insulation

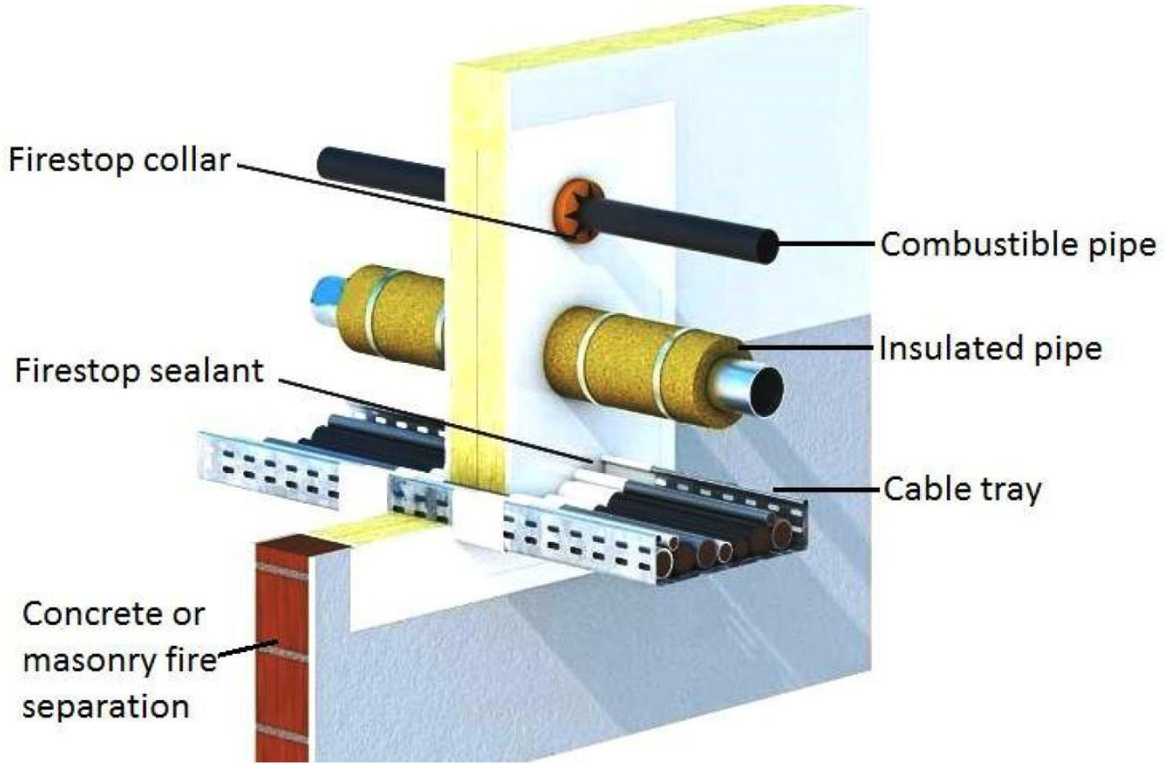


Figure 5: Through penetration Firestop System

Interior Finishes

In accordance with NFPA 101 Section 10.2.3.4 and NFPA 255, Products required to be tested in accordance with ASTM E84, Standard Test Method for Surface Burning Characteristics of Building Materials, or ANSI/ UL 723, Standard for Test for Surface Burning Characteristics of Building Materials shall be classified as follows in accordance with their flame spread and smoke development:

1. Class A interior wall and ceiling finish are characterized by the following:
- Flame spread, 0–25
- Smoke development, 0–450

No continued propagation of fire in any element thereof when so tested

2. Class B interior wall and ceiling finish are characterized by the following:
- Flame spread, 26–75
- Smoke development, 0–450

3. Class C interior wall and ceiling finish are characterized by the following:
- Flame spread, 76–200
- Smoke development, 0–450

Interior floor finishes shall be classified based on test results from NFPA 253, Standard Method of Tests for Critical Radiant Flux of Floor Covering Systems Using a Radiant Heat Source or ASTM E 648 Standard Test Method for Critical Radiant Flux of Floor Covering Systems Using a Radiant Heat Energy Source.

(1) Class I interior floor finish shall be characterized by a critical radiant flux not less than 0.45 W/cm2.
(2) Class II interior floor finish shall be characterized by a critical radiant flux not less than 0.22 W/cm2 but less than 0.45 W/cm2.

Based on the requirements of UAE Fire Code Table 1.12 and with the provision of an automatic sprinkler system throughout the proposed project, the interior wall, ceiling and floor finish shall be as follows:

Walls and Ceilings - Interior Finish				
Occupancy	**Exits**	**Access Corridors**	**Other Spaces**	**Ceiling**
Residential	A or B	A, B or C	A, B or C	A, B or C
Assembly	A or B	A, B or C	A, B or C	A, B or C
Storage	A, B or C	A, B or C	A, B or C	A, B or C

Table 7: Walls, Ceilings and Floors - Interior Finish

Occupancy	**Exits**	**Access Corridors**	**Other Areas**
Residential	I or II	I or II	I or II
Assembly	I or II	I or II	I or II
Storage	I or II	I or II	I or II

Table 8: Interior Floor Finishes

It should be noted that DCD inspectors at the time of inspection are requesting one hour fire resistance rating test certificate for any wooden panel installed in exit corridor or public corridors/ lobbies. The same requirement applies to FHC wooden doors, if any are installed. A 90-minute fire-rated door will be required.

Mezzanine

In accordance with Chapter 1, Table 1.9, UAE Fire Code 2028, following shall be noted:

i. Mezzanines construction shall be with 1 hour fire resistance rating. The aggregate area of mezzanines within a room, other than those located in special-purpose industrial occupancies, shall not exceed one-third the open area of the room in which the mezzanines are located.
ii. Enclosed space shall not be included in a determination of the size of the room in which the mezzanine is located.
iii. There shall be no limit on the number of mezzanines in a room.
iv. the area of the mezzanines shall not be included in the area of the room.
v. All portions of a mezzanine, other than walls not more than 1065 mm high, columns, and posts, shall be open to an unobstructed exit from the room in which the mezzanine is located, unless the occupant load of the aggregate area of the enclosed space does not exceed 10.
vi. A mezzanine having two or more means of egress shall not be required to open into the room in which it is located if not less than one of the means of egress provides direct access from the enclosed area to an exit at the mezzanine level.

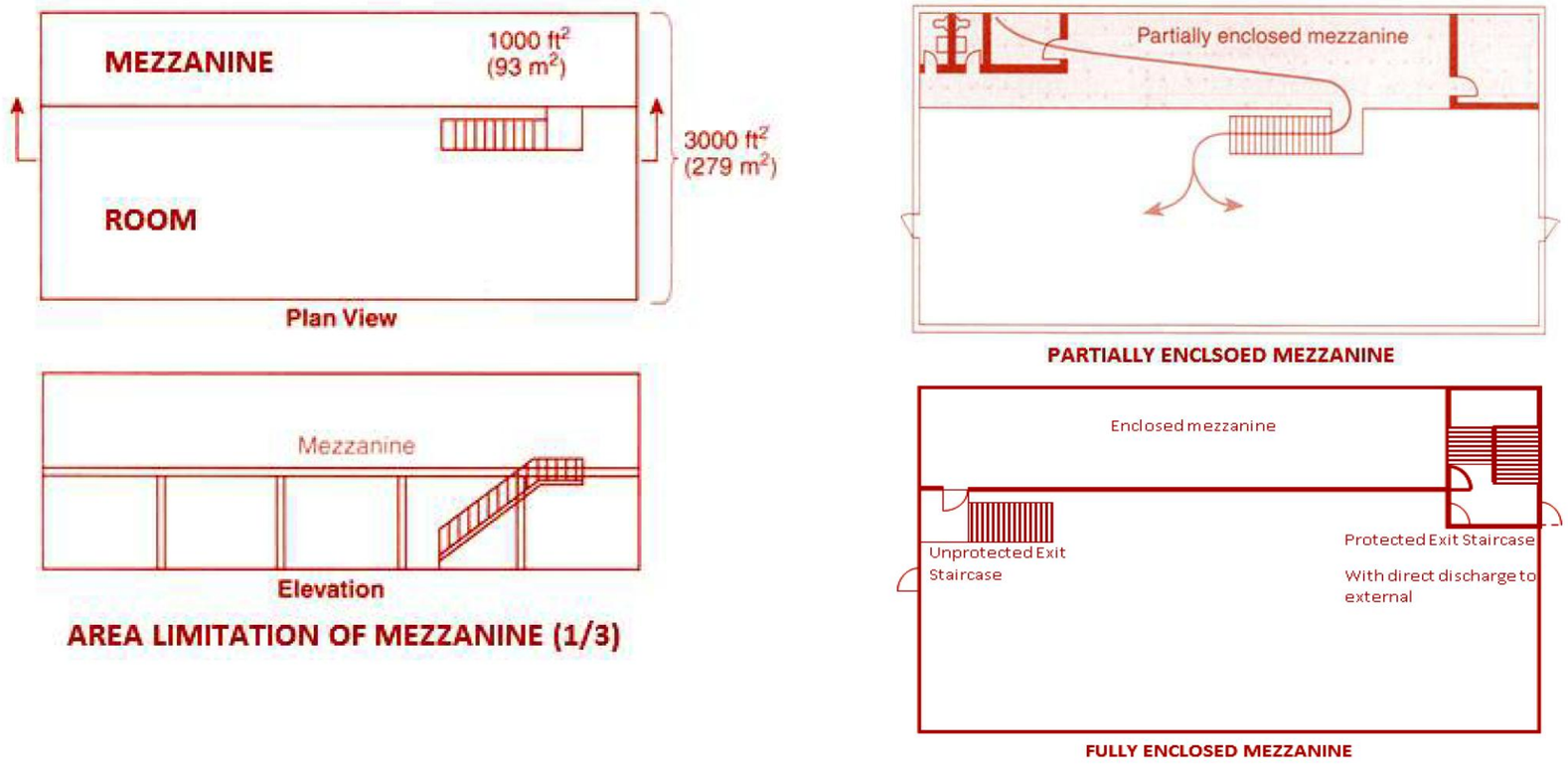


Figure 6: Definition of Mezzanine

MEANS OF EGRESS

In accordance with UAE Fire Code, Chapter 3 Section 1.1, a means of egress is a continuous and unobstructed way of travel from any point in a building or structure to a public way consisting of three separate and distinct parts. Means of egress is expanded to comprise of three separate and distinct parts:

- Exit access is a portion of means of egress that leads to an exit.
- Exit is a portion of means of egress that is separate from all other spaces of a building to provide a protected way of travel to the exit discharge.
- Exit discharge is a part of means of egress between termination of an exit and public way.

Number of Means of Egress

Accordance with Chapter 3, Table 3.14 of UAE Fire Code, each floor level or room is required to be provided with sufficient number of exit width/exit capacity to accommodate the occupant load. The minimum required numbers of exits are listed in the Table below.

Item	Minimum Required Number of Exit
Every floor	2 Exits
Separate exit accessible from every part of every floor	2 Exits
More than 500 but not more than 1000 occupants	3 Exits
More than 1000 occupants	4 Exits

Table 9: Exits Requirement

Remoteness between two exits

Exit Separation shall comply with Table 3.15.a. Chapter 3 of UAE Fire code as reproduced below:

Building	Exit Remoteness	Measurement of Distance
1. Low-rise Sprinklered	1/3 of floor diagonal	Shortest walking distance between exits
2. Low-rise, Non Sprinklered	1/2 of floor diagonal	Shortest walking distance between exits
3. Midrise, Sprinklered	1/3 of floor diagonal	Shortest walking distance between exits
4. Midrise, Non Sprinklered	1/2 of floor diagonal	Shortest walking distance between exits
5. Highrise, Sprinklered	1/3 of floor diagonal	Shortest walking distance between exits

Table 10: Exit Separation



Figure 7: Exit Separation Highrise

For the proposed high-rise sprinklered project, the minimum exit separation of one-third (1/3) of the floor diagonal will be complied with at all levels. Further details are provided in FLS drawings.

Exit Discharge

In accordance with Chapter 3, Table 3.12, UAE Fire Code, exit discharge is providing occupants with a safe path of travel from exit to public way and should comply with the requirements below:

- For sprinkler protected buildings, minimum of 50% of the required number of exits, and minimum of 50 % of the required egress capacity, should discharge directly to the outside of the building through yards, courts, open spaces, or similar spaces to open to sky, leading to public way or directly to a public way.
- In sprinklered buildings, not more than 50% of the required number of exits, and not more than 50% of required exit capacities are permitted to discharge through interior building areas such as lobbies and corridors.
- The interior exit discharge should lead to a free and unobstructed way to the exterior of the building, and such way should be readily visible and identifiable from the point of discharge from the exit.
- Direct discharge should be directly towards the public way and not at the back of the building where occupants still need to travel across the building exterior to reach point of safety of public way.

Impediments to Exit Discharge

Arrangement of the means of egress should be in accordance with the following requirements:

- Access to an exit should not pass-through kitchens, store rooms, restrooms, closets, bedrooms or similar spaces or other rooms or spaces subject to locking.
- Exit access and exit doors should be designed and arranged to be clearly recognizable.
- Corridors should provide exit access without passing through any intervening rooms other than corridors, lobbies and other spaces permitted to be open to corridor.

Stair Landings

In accordance with Chapter 3, Table 3.4, UAE Fire Code, stairs and intermediate landings should continue with no decrease in width along the direction of egress travel. In buildings, every landing should have a dimension, measured in the direction of travel that is not less than the width of the stair. The maximum permitted height of each flight of stairs between landings is 3660mm. Stair landings should be solid, without perforations. Stairs should be constructed of RCC (reinforced concrete).

The clearance between the leading edge of the opening door leaf and stair newel post should be required not less than half of the minimum required width of the exit stair refer to the graphics that follows.

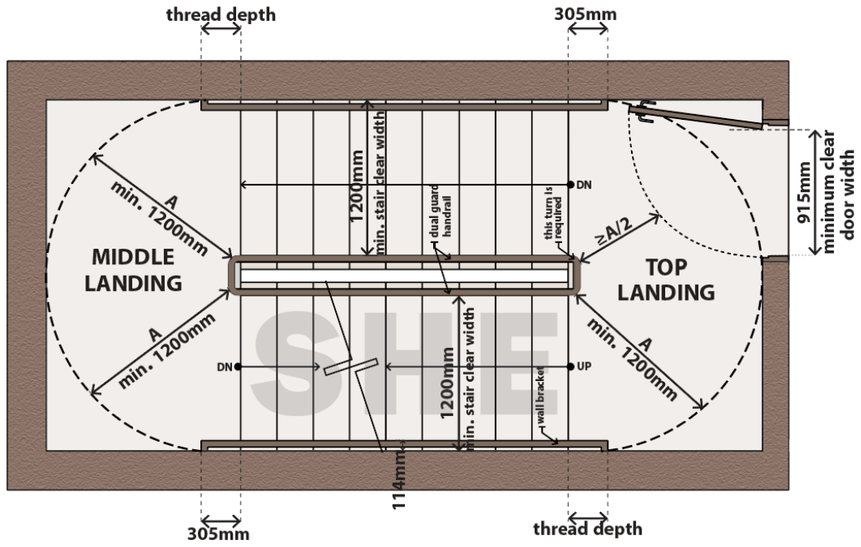


Figure 8: Maximum Permitted Landing Reduction

Stair Riser and Tread Depth

In accordance with Chapter 3, Table 3.2, UAE Fire Code, the rise of steps should be required to be between 100mm and 180mm. The minimum tread depth should be required to be 280mm. The riser height should be required to be measured between two subsequent leading edges of steps. Stair tread depth should be solid, without perforations.

In accordance with Chapter 7, Section 7.2.2.3.6, NFPA 101, variation in excess of 4.8mm in the depth of adjacent treads or in the height of adjacent risers should be prohibited. The tolerance between the largest and smallest riser or between the largest and smallest tread should not exceed 9.5mm in any flight.

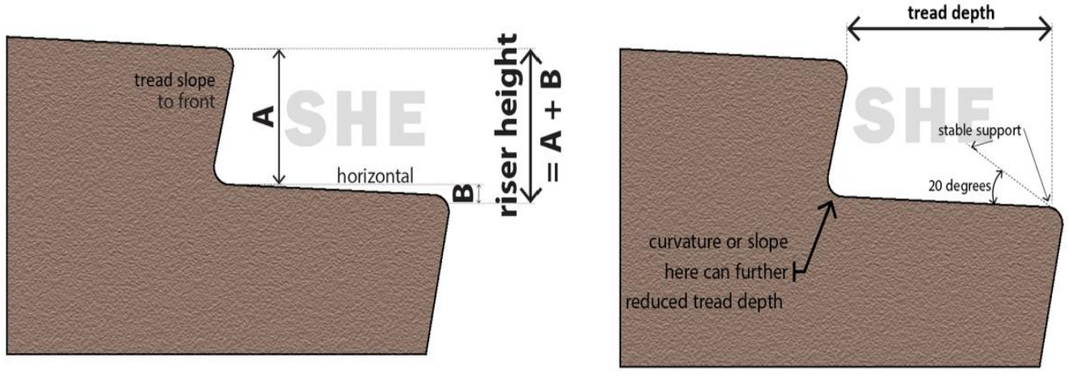


Figure 9: Measurement of Riser and Tread depth

Stair Headroom Clearance

In accordance with Chapter 3, Table 3.4, UAE Fire Code, the minimum clear height inside staircases, above steps, should be required to be 2030mm measured vertically above a plane parallel to, and tangent with, the most forward projection of the stair tread with a tolerance of 19mm, above the finished floor. In other components of means of egress such as corridors and exit passageways the headroom should not be less than 2285mm, with projections from the ceiling not less than 2030mm. nominal above the finished floor of steps.

The minimum ceiling height should be maintained for not less than two-thirds of the ceiling area of any room or space, provided that the ceiling height of the remaining ceiling area is not less than 2030 mm.

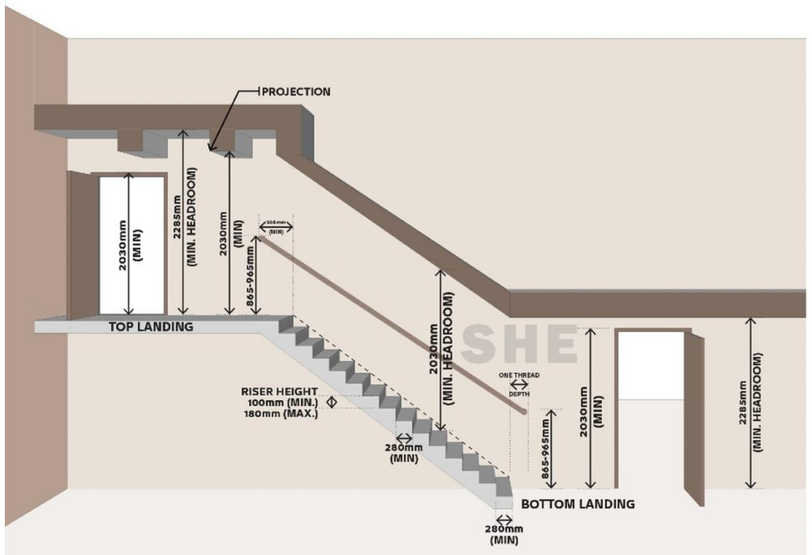


Figure 10: Measurement of Headroom Clearance

Handrails

In accordance with Chapter 3, Table 3.4, UAE Fire Code, Stairs and Ramps should have handrails on both sides. Handrails should be provided within 760mm of all portions of the required egress width. Handrails on stairs and ramps should not be permitted to be less than 865mm and not more than 965mm above the surface of the tread, measured vertically from a line drawn through the outside edges of the nosing or from the surface of a ramp. The maximum projections of handrails allowed in the required width is 100 mm on each side.

The height of required handrails that form part of a guard should be permitted to exceed 965mm, but should not exceed 1065mm, measured vertically to the top of the rail from the leading edge of the tread.

Handrails should be installed to provide a clearance of not less than 57mm between the handrail and the wall to which it is fastened.

Handrails should be required to have a circular cross section with an outside diameter of not less than 32mm and not more than 51mm or a shape that is other than circular with a perimeter dimension of not less than 100mm, but not more than 160mm, and with the largest cross-sectional dimension not more than 57mm, provided that graspable edges are rounded so as to provide a radius of not less than 3.2mm.

Handrails that are not continuous between flights should be extended horizontally 305 mm outside the top riser and continue to slope for a depth of one tread outside the bottom riser. Handrails ends should be returned to the wall or floor or should terminate at newel posts.

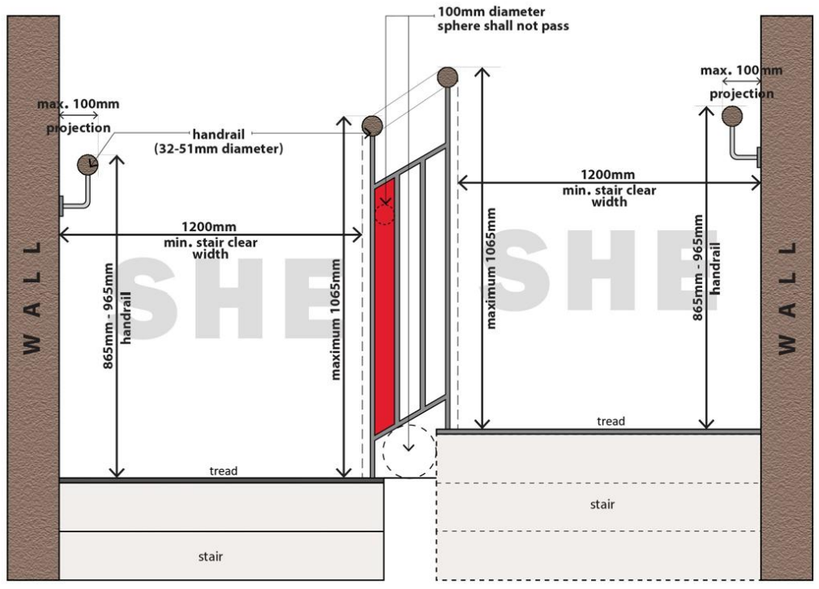


Figure 11: Handrail and Stair Width Measurement

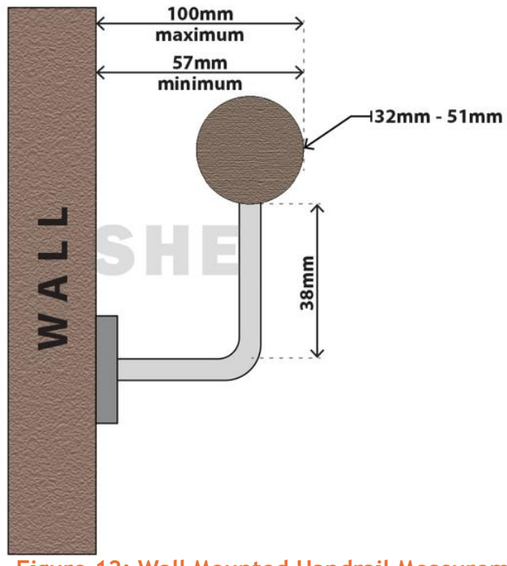


Figure 12: Wall Mounted Handrail Measurement

Guards

In accordance with UAE Fire Code, Chapter 3 Table 3.1, guard requirements are as follows:

- Guards should be provided for every elevated means of egress, open landing for stairs where elevated surface is more than 760mm from finished ground level.
- Such guards should not encroach the required clear width of the stair.
- The height of guards should be measured vertically to the top of the guard from the surface adjacent thereto.
- Guards should be not less than 1200 mm high. In case of stair or ramp handrails that form part of a guard, the height of the guard can be reduced to 1065 mm.
- Open guards should have intermediate rails or an ornamental pattern up to a height of 865 mm, such that a sphere 100mm in diameter is not able to pass through any opening.
- The triangular openings formed by the riser, tread, and bottom element of a guardrail at the open side of a stair should be of such size that a sphere 150mm in diameter is not able to pass through the triangular opening.

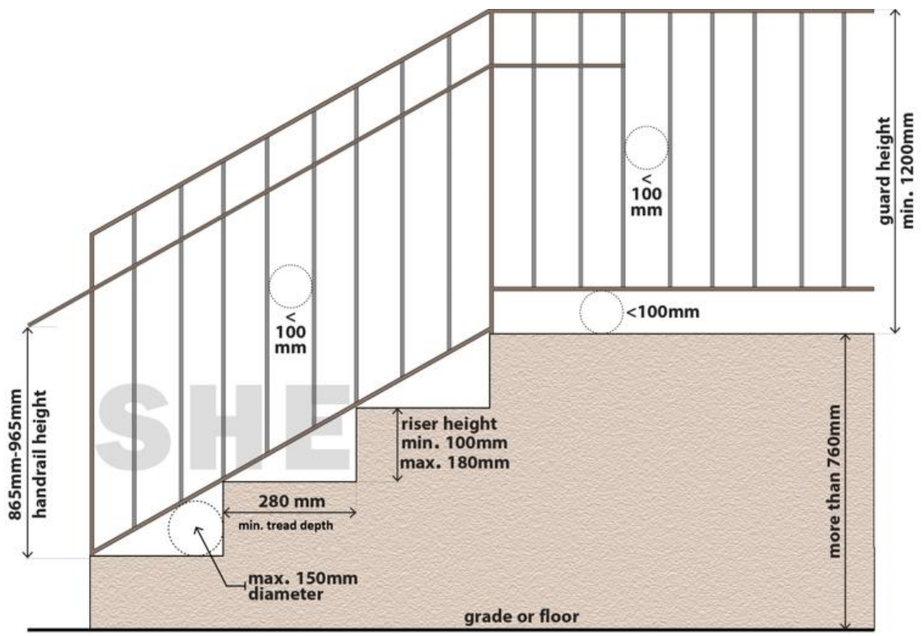


Figure 13: Measurement of Guard.

Corridors

In accordance with Chapter 3, Table 3.24 UAE Fire Code, the residential floors; corridors shall have a minimum 1-hour fire resistance rating. In addition, grilles and louvers shall be prohibited in walls or doors of exit access corridors.

The egress capacity of the corridor shall not be less than the required capacity of the exit to which the corridor leads. In accordance with Chapter 3, Table 3.24 UAE Fire Code, corridor width is tabulated below:

Corridor Width	
Occupancy	**Required Width**
Residential, Assembly	1200 mm

Table 11:Corridor Width

Ramps

In Accordance with UAE Fire Code, Table 3.9:

- All ramps serving as required means of egress shall be of permanent fixed non-combustible construction.
- The ramp floor and landings shall be solid and without perforations.
- Ramps shall have landings located at the top, at the bottom, and at doors opening onto the ramp.
- Every landing shall have a width not less than the width of the ramp.
- Where the ramp is not part of an accessible route, the ramp landings shall not be required to exceed 1220 mm in the direction of travel, provided that the ramp has a straight run.
- Any changes in travel direction shall be made only at landings.

- Ramps and intermediate landings shall continue with no decrease in width along the direction of egress travel.

Ramp Details	
Feature	**Dimensional Criteria**
Minimum unobstructed width	1200 mm
Minimum projections at or below handrail height on each side	114 mm
Maximum Ramp Slope	1 in 12
Maximum Cross Ramp Slope	1 in 48
Maximum Rise for Single Ramp Run	760 mm
Minimum width of landing in direction of travel	1200 mm

Table 12: Ramp Details

Change in Level

In accordance with Table 3.1 of UAE Fire Code, requirements are as follows:

- Abrupt changes in elevation of walking surfaces should not exceed 6.3 mm.
- Changes in elevation above 6.3 mm to 13 mm, should be beveled with a slope of 1 in 2.
- Changes in elevation exceeding 13 mm but not in excess of 535 mm shall be considered a change in level and should be achieved by either a ramp with maximum slope of 1 in 12 or stair with tread depth not less than 330 mm.
- Changes in level in means of egress should be achieved by an approved means of egress where the elevation difference exceeds 535 mm.

BALCONY TERRACE AND WINDOW

Minimum Construction Requirements for Balconies and Railings

In accordance with UAE Fire Code, Chapter 1, Section 2.17.2, Guards are required for any space, walking or standing which is elevated 760 mm above the finished grade level.
From the finished floor level, height of the balcony, terrace railings, elevated space railings should not be less than 1200 mm.
For balusters openings, the separation distance between vertical posts or members of balcony, guardrail or handrail shall not be more than 100 mm.
Any opening formed by either vertical posts, curved frames or design features in the balcony or railings shall not be more than 100 mm.
Horizontal elements, bars, climbable features shall not be installed up to 760 mm.
If the design demands any gap between finished floor surface and the bottom most horizontal component of the railing, such gap shall not be more than 100 mm.
The balcony, Handrail, Guard assembly should be able to withstand a single concentrated load of 200 pounds (0.89 KN), applied in any direction at any point.
If the design demands usage of glass panels in the balcony construction, such glass shall be reinforced glass which can withstand a load of 200 pounds (0.89kN), applied in any direction at any point.
Balconies and terraces should not have accesses with self-closing or self-latching doors, which can accidentally lock people outside in the balcony or terrace, compelling them to misadventures such as climbing, descending or crossing over to other balconies.



Figure 14: Balcony Handrail Details

Minimum Construction Requirements for Windows

In accordance with UAE Fire Code, Chapter 1, Section 2.17.3, Openable Windows shall open outward from the top. Openable windows shall not open from below.

Sliding and openable Windows shall not be located at (F), a height less than 865 mm from the finished floor surface level. If window base is present on the finished floor, the height (F) shall be measured from the top of such window base. See Figure 15a. The window pane, if openable, shall not create a gap of more than 100 mm from the wall, when opened from the top.

The window assembly, shall have reinforced or safety glass and shall be able to withstand a single concentrated load of 200 pounds (0.89 kN), applied in any direction at any point.

There should not be any construction features at the bottom of the window which can be exploited into ‘ladder or climbing feature’. Provide locks and safety measures for sliding windows. **See Figure 15b.**



Figure 15: Window specifications



Figure 15b: Window Safety

DOOR

Doors

In accordance with Chapter 3, Table 3.2, UAE Fire Code, every door and door assembly shall be required to be designed and constructed so that the way of egress travel is obvious and direct.
In accordance with Chapter 3, Table 3.2, the UAE Fire Code, doors opening in means of egress will be required to be not less than 915 mm in clear width. Where a pair of doors is provided, not less than one of the doors should have 810 mm clear width. Exit access door assemblies serving a room not exceeding 6.5m2 and not required to be accessible to persons with severe mobility impairments shall be not less than 610mm in door leaf width.

The door assembly shall be readily operable from the egress side without special knowledge or effort. The forces required to fully open any door leaf manually in a means of egress shall not exceed:

- 67 N (15 lbf) to release the latch,
- 133 N (30 lbf) to set any door leaf in motion,
- 67 N (15 lbf) to open any door leaf to the minimum required width

For swinging door assemblies, clear width will be required to be measured as follows:

- The measurement should be taken at the narrowest point in the door opening.
- The measurement should be taken between the face of the door leaf and the stop of the frame. For swinging door assemblies, the measurement should be taken with the door leaf open 90 degrees.
- Projections of not more than 100mm into the door opening width on the hinge side should not be considered reductions in clear width, provided that such projections are for purposes of accommodating panic hardware or fire exit hardware and are located not less than 865mm and not more than 1220mm above the floor.
- Projections exceeding 2030mm above the floor should not be considered reductions in clear width.

The elevation of the floor surfaces on both sides of a door opening shall not vary by more than 13mm and it should be maintained so for not less than width of widest door leaf. Thresholds at door openings should not exceed 13mm in height. Where doors serve spaces that are not normally occupied, the floor level will be permitted to be lower than that of the door opening but should be not more than 205mm lower.
Balcony and terrace access shall be restricted with door having child lock or latch. Window opening should be restricted to 100mm opening with windows safety latch. Every closet door latch should be such that children can open the door from inside the closet. Every bathroom door should be designed to allow opening from the outside during an emergency when locked. At all times, all locking devices that impede or prohibit egress or that cannot be easily disengaged shall be prohibited.

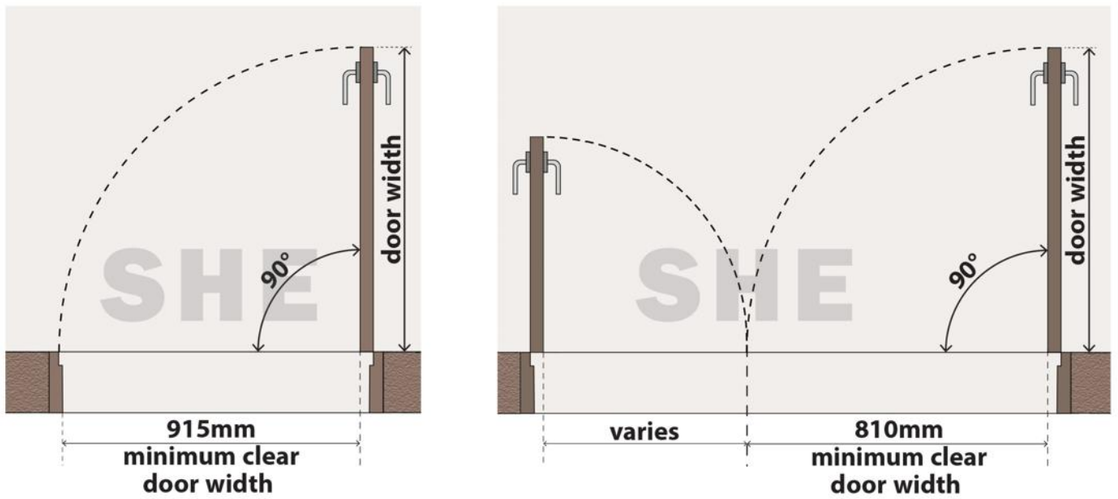


Figure 16: Minimum Door width

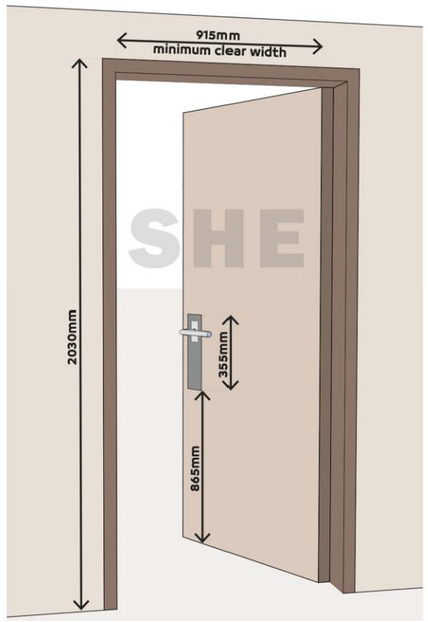


Figure 17: Minimum Door width

Door Swing

In accordance with Chapter 3, Table 3.2 UAE Fire Code, all doors in a means of egress shall be of the side hinged or pivoted swinging type and should be installed to be capable of swinging from any position to the full required width of the opening in which it is installed. Doors will be required to swing in the direction of egress travel were serving a room or area with an occupant load of 50 or more or serving horizontal exit or serving high hazard content room or serving an exit enclosure.

Doors serving Residential Units shall not be required to swing in the direction of egress.

During its swing, any door in a means of egress shall leave not less than one-half of the required width of an aisle, a corridor, a passageway, or a landing unobstructed and should project not more than 180 mm into the required width of an aisle, a corridor, a passageway, or a landing, when fully open. Refer to figure below:

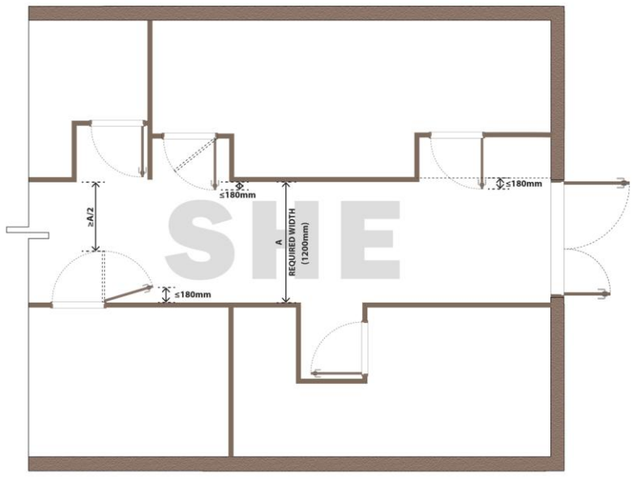


Figure 18: Door leaf swings into corridor

Self-Closing and Automatic Closing Devices

In accordance with Chapter 3, Table 3.2, UAE Fire Code, doors normally required to be kept closed shall not be secured in the open position at any time and shall be self-closing or automatic closing. Doors should be permitted to be provided with automatic-closing/hold-open devices, where the following conditions apply:

1. Upon release of the hold-open mechanism, the door becomes self-closing.
2. The release device is designed so that the door instantly releases manually and upon release, becomes self-closing.
3. The automatic releasing mechanism or medium is activated by the operation of approved smoke detectors.
4. Upon loss of power to the hold-open device, the hold-open mechanism is released, and the door becomes self-closing.
5. The release by means of smoke detection of one door in a stair enclosure results in closing all doors serving that stair.

In accordance with Table 3.3a UAE Fire Code, self-closing door is required on the following locations:

- Garbage room
- Elevator Lobby
- Exit stairs
- All fire rated doors

OCCUPANT LOAD/ EXIT WIDTH

In accordance with Table 3.4 of UAE Fire Code, minimum stair dimension shall be provided as follows:

Stair Dimension Requirements	
Feature	**Dimensional Criteria**
Minimum Stair and landing width < 2000 occupants	1200mm
Minimum Stair and landing width > 2000 occupants	1420mm

Table 13: Stair Dimension Requirements

The required width of a stair should be measured from wall to the face of handrail. The maximum projections of handrails allowed in this required width is 100 mm on each side, at a height of 865 mm – 965 mm. Stairs should be constructed of RCC (reinforced concrete).

In accordance with UAE Fire Code, Chapter 3 Table 3.13, the required Egress Capacity is calculated based on the Occupant Load Factor provided as follows:

Occupant Load Factor	
Use	**m^2/ person**
Apartment units	18.6
Assembly Use	
Fitness Room (Exercise room with Equipment)	4.6
Swimming Pool (Water Surface)	4.6
Pool Deck	2.8
Less Concentrated	1.4 net
Storage	27.9
Offices	9.3

Table 14: Occupant Load Factors

The egress capacity shall be calculated based on its use and the calculated occupant load.

Capacity Factor	
Stairways (width/ person)	**Level Components and Ramps (width/ person)**
7.6 mm	5.0 mm

Table 15: Capacity Factor

The total anticipated (calculated) occupant load for the proposed project is attached as appendix A to this report.

TRAVEL DISTANCE

Maximum Travel distances are measured from the most remote point of the floor to a nearest fire protected fire exit. Travel distance requirements are directly in relation to the different occupancy classification and use. There are three (3) travel distance criteria as part of the means of egress design, as follows;

- Common Path of Travel
- Dead-end corridor
- Travel Distance

The following are descriptions of the above components:

Common Path of Travel

As a general principle, a common path of travel exists where a space is arranged so that occupants within that space are able to travel in only one direction to reach any of the exits or to reach the point at which the occupants have the choice of two paths of travel to remote exits, see figure below:
This common path limit applies to areas within all spaces, up to a point where an occupant can go in two directions.
Common path does not apply inside residential units.

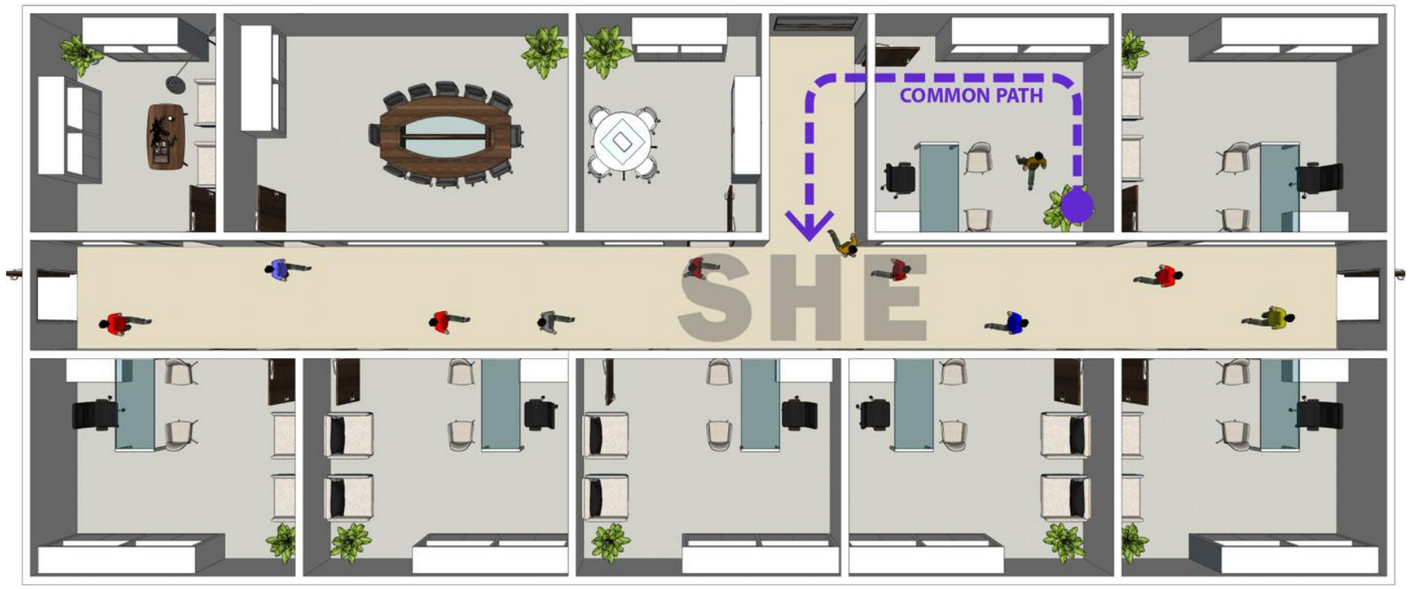


Figure 19: Common path of travel

Dead-end

While a dead end is similar to a common path of travel, a dead end can exist where there is no path of travel from an occupied space but can also exist where an occupant enters a corridor thinking there is an exit at the end and, finding none, is forced to retrace his or her path to reach a choice of exits see figure below:



Figure 20: Dead end

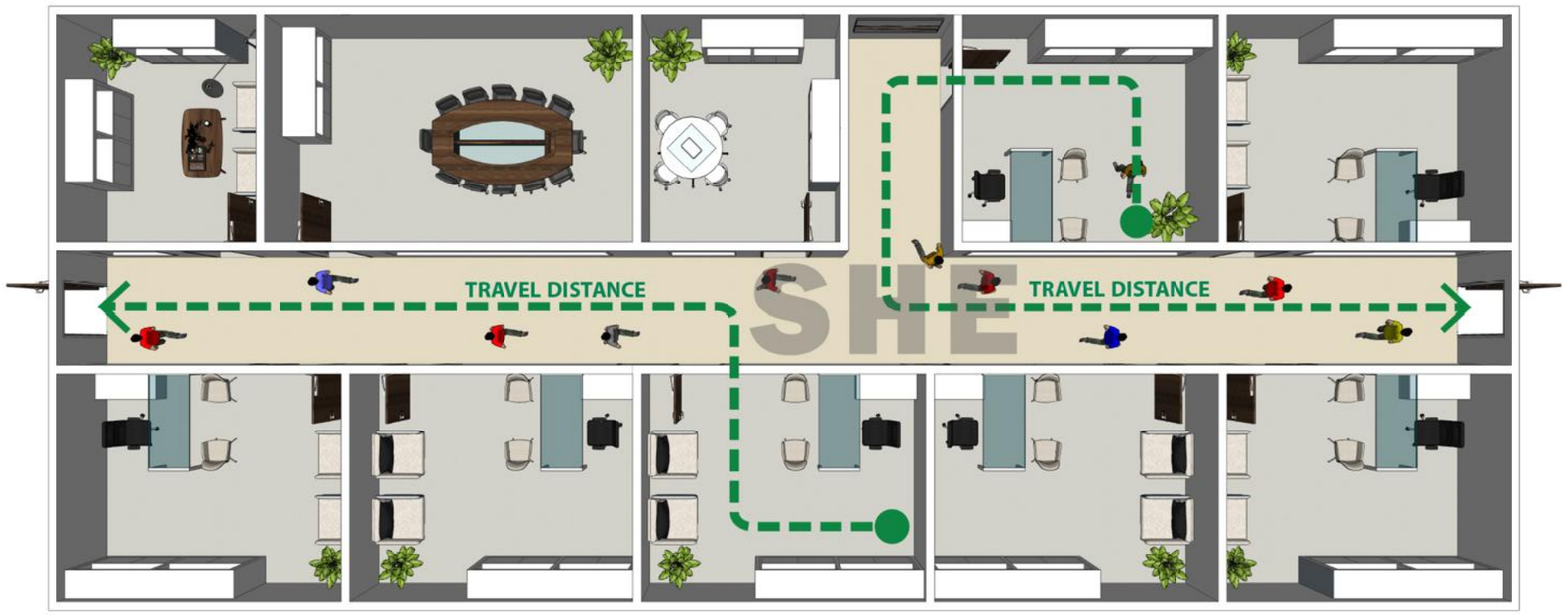


Figure 21: Travel Distance

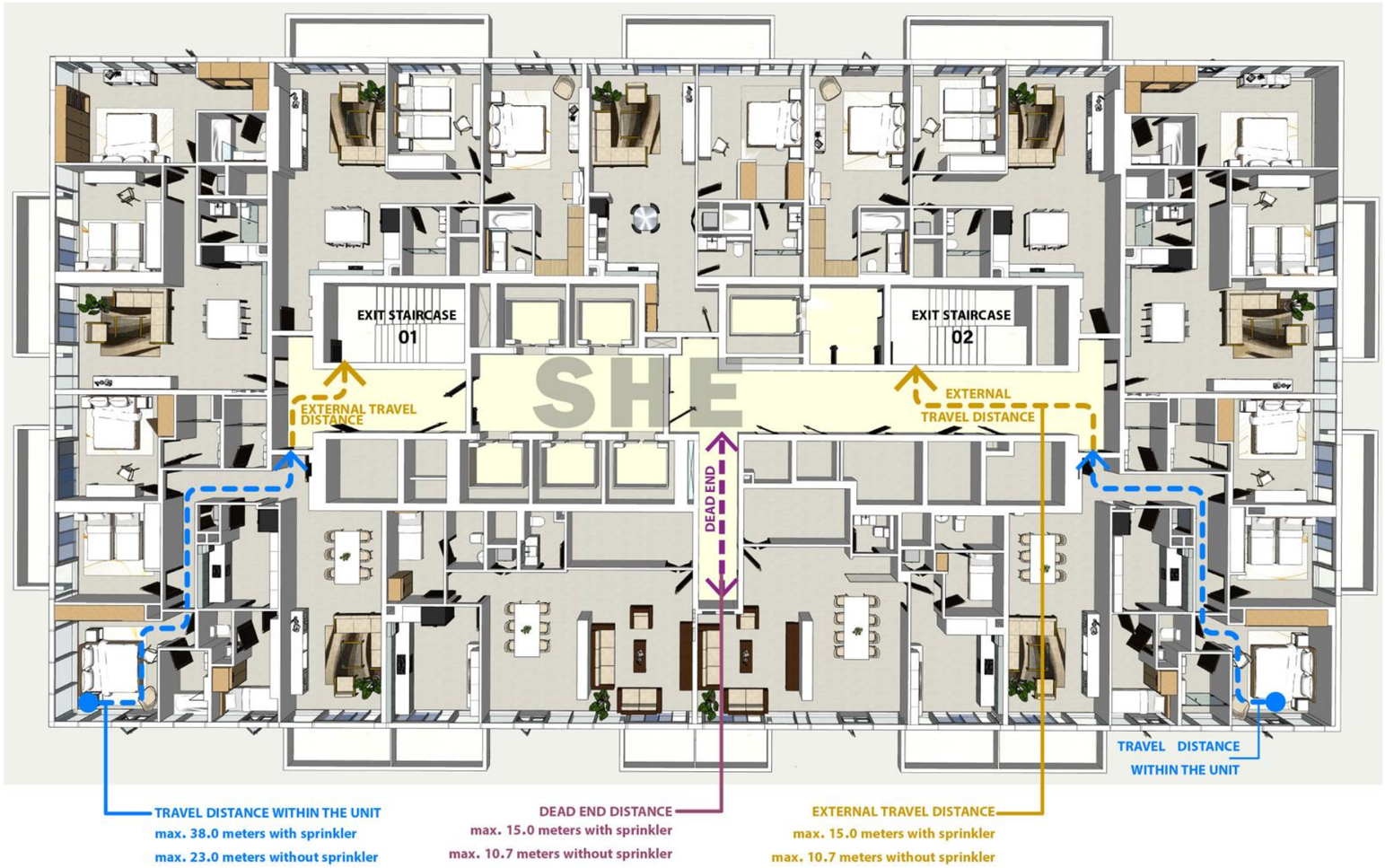


Figure 22: Travel Distance inside units

In accordance with Chapter 3, Table 3.16, UAE Fire Code, the maximum permitted travel distances for this project are tabulated below:

Type of Occupancy	Distance
Residential	
Apartment Units	
Travel Distance (from Apartment door to an exit stair)	61 m
Travel Distance inside the unit	38 m
Dead-end Limit	15 m
Assembly Occupancy	
Travel Distance	76 m
Maximum Common Path (<50 Persons)	23 m
Maximum Common Path (>50 Persons)	6.1 m
Dead-end Limit	6.1 m

Table 16:Maximum permitted Travel Distances, Common Path, Dead-end

For the proposed project, except for a few parking/landscape areas which are currently under coordination with the Architectural team the travel distance requirements are complying in general, the final arrangement will be reflected during the Developed Design at the next stage (refer to FLS drawings).

FIRE FIGHTING ACCESS

General Requirements for High-rise Building

- As the project is categorized as Highrise buildings, therefore in accordance with Table 2.5 Chapter 2 of UAE Fire code, the extend of fire engine access-way shall be as illustrated in table below:

Extent of Fire Vehicle Access for High-rise Building (with Sprinklers)	
High-rise Buildings	**Extent of Fire Access**
Less than 4000 m^2	Minimum of 1/6 perimeter (at least 15 meters)
4001 m^2 to 8000 m^2	1/4 perimeter
8001 m^2 to 16,000 m^2	1/2 perimeter
16,000 m^2 to 32,000 m^2	3/4 perimeter
More than 32,000 m^2	Whole perimeter (100%)

Table 17: Extent of the fire vehicle for High-rise Building

The proposed project provides fire truck access along 100% of the building perimeter. The access road is located not less than 2 meters and not more than 10 meters from the building façade (except for one area on the right side which will be coordinated with the Architectural Team at the next stage). The access road is 6 meters wide and shall be maintained free of any obstructions, including trees, fixed furniture, landscaping elements, or other permanent structures. Furthermore, no canopies, balconies, or other overhead projections shall encroach upon the required emergency access route.

Fire Truck Access (FTA) is provided around the entire plot; however, in order to reach the building perimeter within the required 2–10 meter distance from the façade, the fire truck may need to travel against the direction of the road (in reverse). This arrangement will be subject to approval by Dubai Civil Defence (DCD).

Note: The plot will be surrounded by large trees, palm trees, and tall shrubs of different heights planted at a minimum separation distance of 10 meters, which may create potential obstructions for firefighters during emergency operations. This condition will be subject to review and confirmation by the inspection team.

Fire Access for Fire Vehicle

- The minimum width of the firefighting access-way shall be 6 m throughout and must be able to accommodate the entry and maneuvering of a fire engine, extended ladders, pumping appliances, aerial appliances, turntable and/or hydraulic platforms.
- The fire engine access-way shall be provided within 18 m of the breeching inlet with a maximum parking distance of Civil Defence vehicle from building entrance shall not exceed 15 m.
- The Fire engine access-way shall be located at least 2 m from the building, but not more than 10m away from the external wall or façade of the building.
- The fire engine access-way shall be laid on a level platform or if on an incline, the gradient shall not exceed 10%.
- Public roads can serve as fire engine access provided that the locations of such public roads are in compliance with the fire engine access requirements.
- The access-way and fire engine access roads shall be kept clear of all obstructions and other parts of the building, plants, trees or other fixtures shall not obstruct the path between the access-way and the access openings.
- A suitable access pathway shall also be provided to enable fire-fighters to inspect all elevations of a building during or after a fire. A suitable pathway could be a paved or gravel path. Any such pathway shall be a minimum of 1 m wide and Plants, Bushes, Walls or other features shall not impede such pathway.
- Fire access roadways and fire access way shall be free of obstruction and designated parking for fire vehicles shall be available at all times. Signs shall be provided with ‘NO PARKING, FIRE LANE’ and ‘PARKING RESERVED OF CIVIL DEFENCE’ signage. Size of wordings shall not be less than 50 mm.

Fire Access Roadways

Specification of the fire access roadways and route specifications is outlined in table below:

Access Roadways and Route Specifications		
Item	**For Standard Fire Appliances**	**For High Reach Fire Appliance**
Minimum weight carrying capacity of all-weather driving surface to withstand Civil Defence Vehicle	41,000 kg	81,000 kg
Maximum Dead-end	45 m	45 m
Minimum unobstructed width	6 m	6 m
Minimum unobstructed vertical clearance for any grade	4.5 m	4.5 m
Maximum Road Grade	10%	10%
Minimum width of road between kerbs	6 m	6 m
Minimum width of gateways	4.5 m	4.5 m
Minimum turning circle (roundabout) radius walls	12 m	12 m
Minimum inner turning radius	7 m	7 m
Minimum outer turning radius	13 m	13 m

Table 18: Specification of the Fire Access Roadway and Route Specifications



Figure 23: Maximum Horizontal Distance

Emergency Command Centre (ECC)

For the high-rise buildings emergency command center shall be provided with adequate size to house all the terminals and supervisory/control equipment, etc. of the building's fire protection/detection systems and should have a free working space of at least 19m2 with no dimension less than 3000mm. ECC shall be located at the entrance side of the building and along fire access road in accordance with Chapter 2 Section 2.9.3 of UAE Fire Code.

The ECC must be very close to the fire-fighting lift lobby or at the main entrance lobby. Air conditioning or mechanical ventilation required for the ECC shall be provided with a secondary power supply and should have ductworks independent of any other ductwork serving other parts of the building.

The ECC shall contain the following:

1. Building plan indicating floor plans and detail locations of building exit staircases, means of egress, exit routes, protected corridors and exit access, fire protection systems, fire compartments, fire-fighting equipment and fire department access.
2. Emergency Voice/Alarm communication system unit.
3. Fire Detection and alarm system unit.
4. Annunciator visually indicating the location of the elevators and their operation.
5. Status indicator and control of air handling system.
6. Controls for unlocking electromagnetics or access control stairway doors.
7. Sprinkler valve and water flow detector display panels.
8. Emergency and standby power status indicators.
9. Fire Pump status indicators.
10. Generator supervision devices and manual start and transfer features.
11. Public address system.
12. Controls for smoke management systems.

Fire Fighting Lifts

In accordance with Chapter 2, Section 2.9.4 UAE Fire Code, since the accommodation buildings is classified as a high-rise, it shall be provided with a firefighting lift. The fire-fighting lift shall be required to be provided with emergency power and fire-fighting controls as per ASME A17.1. The fire-fighting lift shall be required to comply with Dubai Municipality requirements. Fireman's lift car shall have a minimum size and arrangement to accommodate ambulance stretcher of 610mm X 2134mm with not less than 127 mm radius corners, in the horizontal, open position.

The fire lift shall have access to every habitable floor above or below the designated floor. The fire lift shaft should be provided with an operational feature that would enable firemen to cancel first or earlier call which has been inadvertently made to the fire lift during an emergency.

All elevators should return to the recall level after fire alarm activation. If a fire is detected at the recall level, the elevator should be recalled to an alternate recall level.

FIRE FIGHTING SYSTEMS

This section of the report identifies the fire-fighting systems required in the proposed building. The required fire suppression systems shall be provided in accordance with NFPA 101, UAE Fire Code and other applicable NFPA standards and codes.

Sprinkler System

In accordance with Table 9.18 of UAE Fire Code, the high-rise building shall be protected throughout the building by an approved automatic sprinkler system. As per UAE Fire Code and NFPA 13 requirements, the sprinkler system shall comply with the following:

The maximum area of any single sprinkler zone should be limited to 4,831 m2 for Light and Ordinary Hazard areas. Where the floor area exceeds this limit, additional water supply risers should be

proposed for every sprinkler zone in each floor within sprinkler alarm check valve assembly and riser isolation valve.
The minimum operating pressure of any sprinkler for determining the water supply requirements should not be less than 0.5 bar. The maximum operating pressure in a sprinkler system should not exceed 12 bar. Electrical rooms may not be sprinklered if the all the following conditions apply:

- The room is dedicated to electrical equipment only.
- Only dry-type electrical equipment is used.
- Equipment is installed in a 2-hour fire-rated enclosure including protection for penetrations.
- No combustible storage is permitted to be stored in the room

To provide adequate fire protection throughout the building, the hazard of the contents within the building should be required to be classified into Light, Ordinary or extra as per NFPA 13. The design density of sprinkler system based on the hazard categories is given below.

HAZARD	DESIGN DENSITY (mm/min) gpm/ft2	AREA OF OPERATION (m2) ft2	WATER DEMAND ONLY FOR SPRINKLERS gpm
1. Light Hazard	(4.1) 0.10	(139) 1500	150
2. Ordinary 1 Hazard	(6.1) 0.15	(139) 1500	225
3. Ordinary 2 Hazard	(8.1) 0.20	(139) 1500	300
4. Extra Hazard Group 1	(12.2) 0.30	(232) 2500	750
5. Extra Hazard Group 2	(16.3) 0.40	(232) 2500	1000

Table 19: Design Density based on Occupancy Hazard

Clean Agent Fire Suppression System

In areas where water-based suppression system may be undesirable due to the potential of damage to equipment, a clean agent fire suppression system should be provided. The system should be an engineered total flooding system designed and installed in accordance with NFPA 2001.

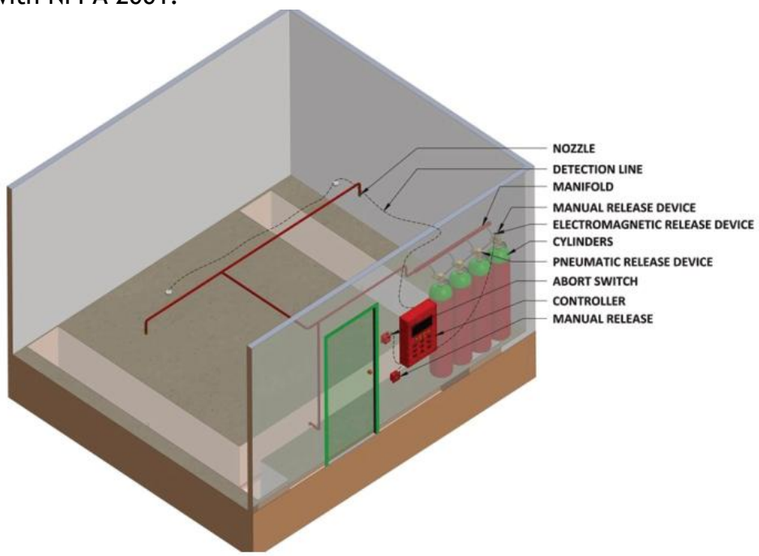


Figure 24: Typical Inert Gas System

Foam Sprinkler System

A deluge foam-water sprinkler system shall be provided for the generator room. A deluge system is a fixed fire protection system in which the piping system is dry until the deluge valve operates to distribute pressurized foam-water from the open nozzles.

The system piping should be hydraulically designed to obtain reasonably uniform foam and water distribution and to allow for loss of head in system piping. A minimum of 3% foam-water solution should be discharged.

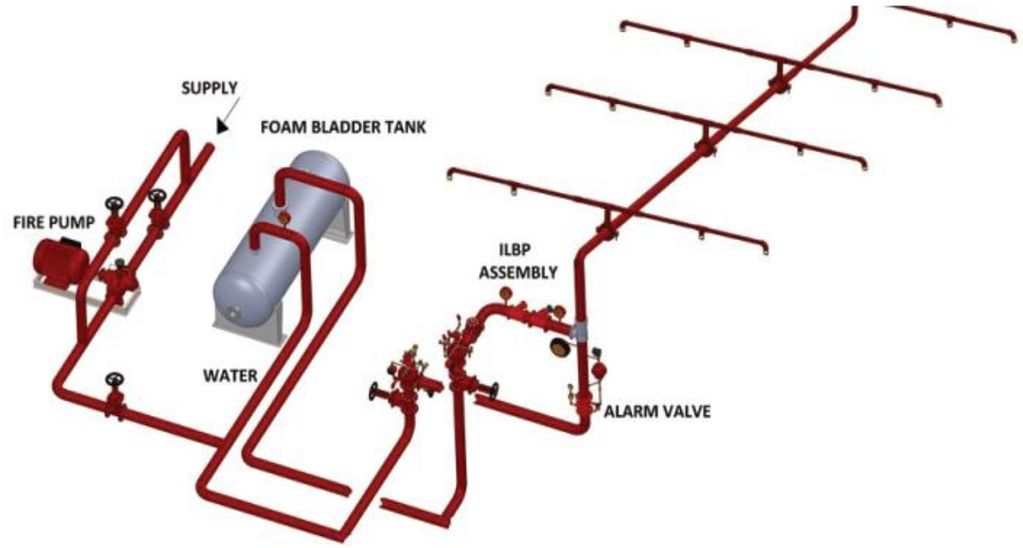


Figure 25: Typical Foam Water Sprinkler System for Diesel Generator Room

Standpipe (Wet Riser) System

In accordance with Chapter 9, Table 9.1 UAE Fire Code, a wet riser system should be required for the entire building.

The wet riser system should be provided with Class III system. A Class III system fire hose station should consist of a 25 mm hose reel or 40 mm fire hose rack & a 65mm diameter landing valve with hoses and nozzles. The hose stations should be located next to each stair enclosures at typical floors and beside each exit way at the ground floor.

Additionally, a fire hose station along the exit access corridor or horizontal exit or in the car park floors should be installed such that all portions of each floor are within 30 m from the fire hose station while measured along the natural path of travel from the hose station.

The wet risers serving this building should be required to be designed in accordance with NFPA 14, “Standard for the Installation of Standpipe and Hose Systems”. The Fire Department connections (breaching inlet) for the sprinkler system and wet riser should be required to be provided along the sides of the building along the fire truck access road.

When most remote landing valve is located at a height 45 m from the fire pump location, the risers should be zoned into “High Zone” and “Low Zone” with looping.

Fire Pumps

Based on UAE Fire Code requirements, the building shall be provided with a dedicated fire pump set to serve the water-based fire protection system. The typical fire pump set consists of a jockey pump, main electric motor-driven fire pump and standby Diesel-engine fire pump. The fire pumps shall be designed, installed and tested for acceptance in accordance with NFPA 20 requirements.

Fire Water Tank

The fire water supply shall be from a permanent water storage tank to cater to the pumping demand. The water tank should be separated into two equal half compartments (50% + 50%) such that

even during water tank cleaning process in one compartment, another compartment is available for the fire protection systems. The fire water tank should be sized to cater the fire-fighting systems for a duration of 60-minutes, complete with automatic low water level detection and instantaneous refilling arrangement.

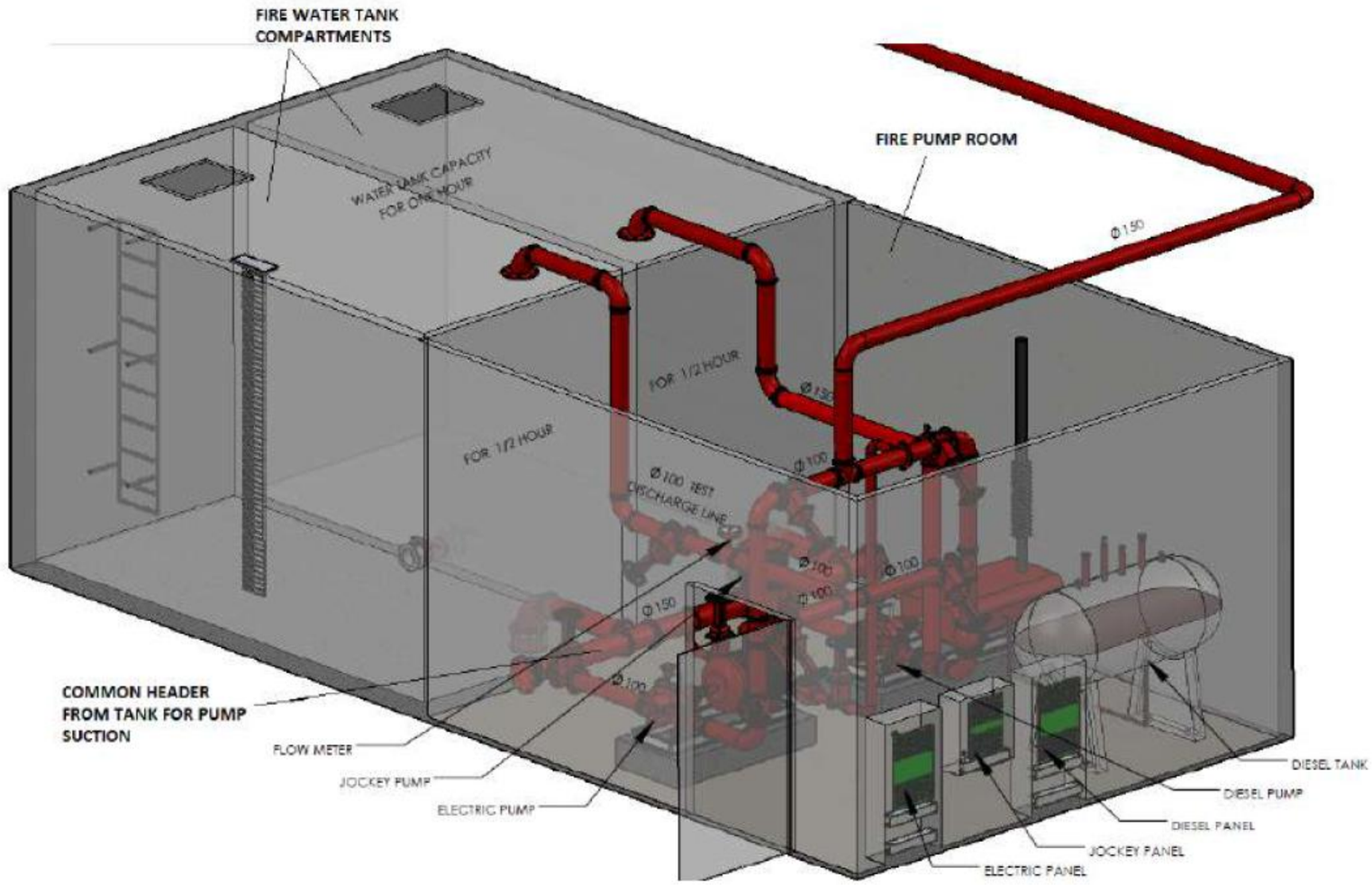


Figure 26: Horizontal Fire Pump for Illustration

Breeching inlets

It is required that the breeching inlets are located on the street side and visible from the fire engine access. A clear path without obstruction to the inlets is required and it is a good practice to provide a walkway with 1200 mm in width from the fire engine access road. Breeching inlets shall be located at a visible main entrance side of the building. The breeching inlet will be required to be within 18m of the fire truck accessway. Breeching inlets shall consist of i. Four 65 mm, (4-Way) direct coupling type fitting for wet riser. ii. Two 65 mm, (2-Way) direct coupling type fitting for Sprinkler Riser. iii. Two 65 mm, (2-Way) direct coupling type fitting for Dry riser. It shall be located not less than 457 mm nor more than 1219 mm above the finished ground level.

Portable Extinguishers

In accordance with Chapter 4, Section 2.1, UAE Fire Code, as a first line of defence during the initial stages of a fire the availability of portable fire extinguishers is mandatory for all occupancies to be used by the occupants to extinguish the fire before the fire grows out of control.

SMOKE MANAGEMENT SYSTEMS

Stair Pressurization

In accordance with Chapter 10 Table 10.4 of UAE Fire Code, for the proposed building as habitable height exceeding 23 m, all exit stairs enclosures serving the building shall be pressurized and meet the requirements for a smoke proof enclosure.

The key criteria for the stair pressurization system are as follows:

- Shall be an approved engineered system.
- Shall achieve the design pressure difference across the barrier with no less than 12.5 Pa for sprinklered building.
- Shall be no more than force of 133 N for the pressure difference across the door.
- The air velocity at the supply air inlets shall not exceed 5 m/s and the air velocity through the open door on the floor of fire shall not be less than 1 m/s.
- The calculation shall consider 2 No. of doors to be opened simultaneously (one door at the floor of fire origin and the discharge door at the grade level).
- Standby fans will not be required.

Lift Pressurization

Lift hoist way pressurization shall be provided for all passenger and fire lifts in high-rise buildings, in accordance with Section 2.19, Chapter 10 of the UAE Fire and Life Safety Code of Practice, 2018.

Corridor Smoke Purging

Corridor and open circulation areas shall be provided with mechanical smoke purge with a mechanical system. The intent of the smoke purge system is for the smoke removal for firefighting operations during and after fire.

The key criteria for corridor smoke purging system are as follows:

- Smoke extract fans should be approved for effective operation at 400°C for 2 hours.
- Corridor smoke purge mechanical system shall be capable of achieving a minimum of 6 air changes per hour.
- Mechanical air supply inlets should achieve a minimum 85% airflow rate of the extract.
- The supply points for the makeup air should be located beneath the smoke layer interface.
- Air supply inlets and smoke extraction outlets should be distributed alternately, such that the horizontal distance between the supply and extraction, measured along the axis of circulation, should not exceed 15 m.
- Extraction outlet shall not be located within 6 meters of any exit stair or exit.
- When a smoke extraction outlet is served by two air supply inlets, the distance between inlets and out let’s must be equivalent.
- Smoke extraction outlets should have their lower part at least 1.830 m above the floor.
- Back-up (Stand-by) fans should not be required for corridor and open circulation area smoke management systems.
- Corridor and open circulation area smoke management system, control panel, automatic smoke curtains, if any, Fans and related equipment should be provided with an emergency power.

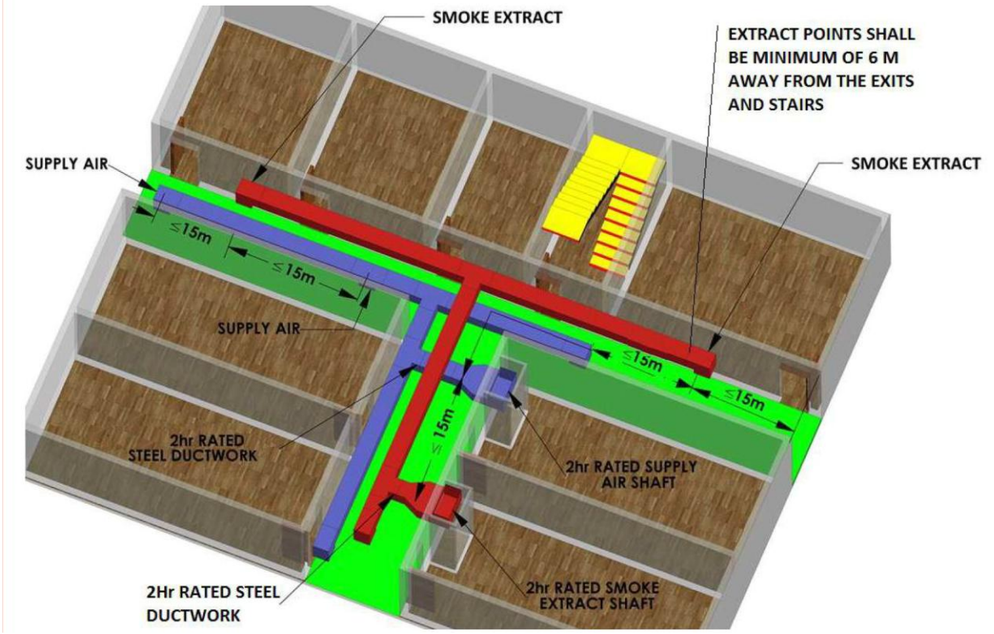


Figure 27: Location of Supply air inlets and Smoke Extract outlets

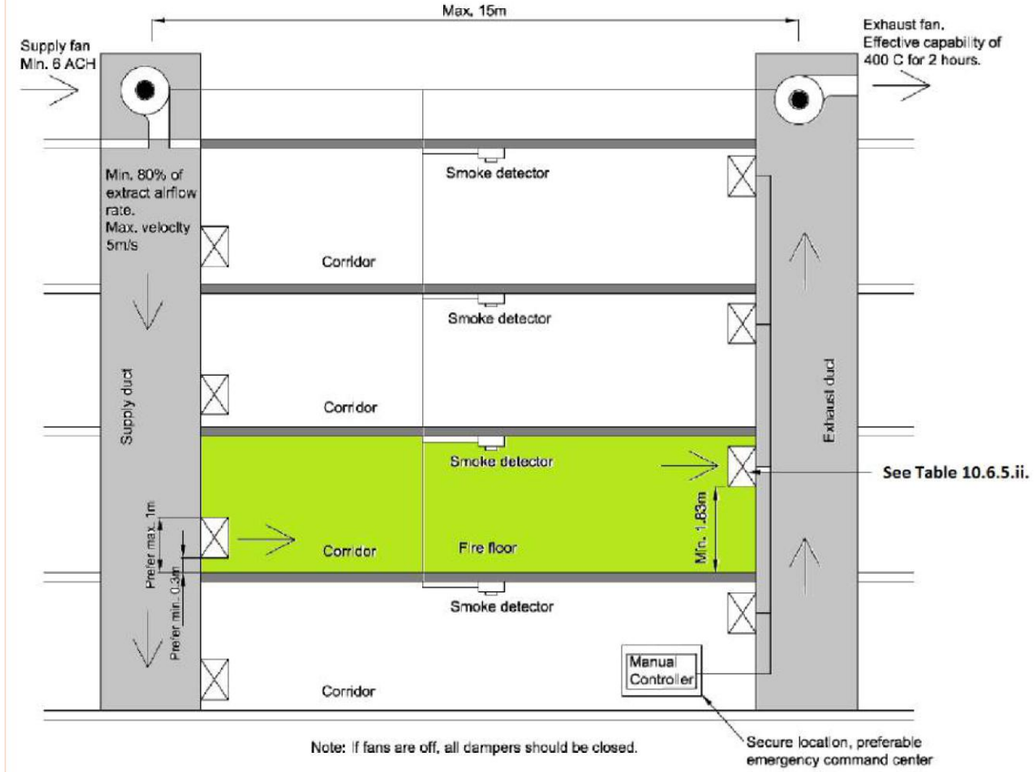


Figure 28: Typical corridor and open circulation area mechanical smoke purge system

Miscellaneous Rooms Smoke Control

The following rooms require a dedicated mechanical ventilation system. Rooms should be temperature controlled dedicated AHU as recommended by the equipment and control panel manufacturers.

- Fire Pump Room
- Generator Room
- Emergency/Fire Command Centre

Mechanical ventilation system should comply with the following requirements:

- The mechanical ventilation system should be an independent system of exhaust mode only exclusive to the particular area.
- Exhaust should not be into underground, basement or car park spaces.
- Exhaust should be directly to the exterior, through dedicated ducts or shafts and should not be less than 5 m from any air intake openings.
- Where the exhaust ducts run outside the area they are serving, they should either be enclosed in a structure or be constructed to give at least the same fire rating as the area they are serving or that of the room through which they traverse, whichever is higher. The rating should apply to fire exposure from both internal and external of the duct or structure.
- Ducts serving systems and areas other than the mechanical ventilated areas should not pass through such rooms.
- Standby power is required for the mechanical ventilation systems serving as smoke exhaust.

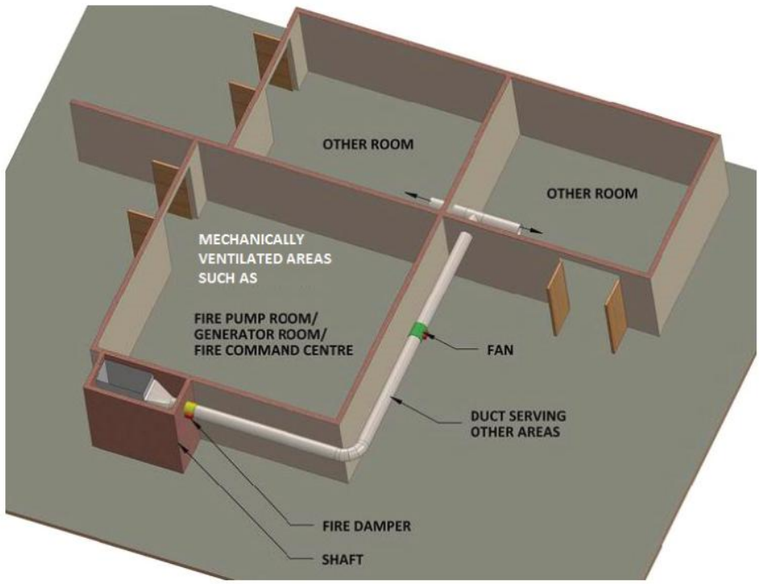


Figure 29: Ducts Serving other Areas than Mechanically Ventilated Areas

LPG SYSTEM

In accordance with UAE Fire Code, LPG distribution shall not pass through the following areas.

- Exit routes, Exit stairs, Exit corridors, Exit lobbies
- Public or assembly areas
- Public seating areas
- Firefighting lobbies, Fire Command Center, Fire pump room, Control Rooms
- Lift shafts, Firefighting riser shafts, Sprinkler valve rooms, garbage chutes, cavity walls, HVAC ducts

- Under building and equipment foundations
- Compartments or ducts dedicated for electrical switchgear, transformers, generators garbage rooms, refrigeration chambers, cold rooms.
- Pipes and vessels containing flammable, oxidizing, corrosive and other hazardous liquids and materials.

As per UAE Fire Code Chapter 11 Table 11.4, LPG Piping Distribution, Filling and Other Requirements are as follows:

- 6 m distance shall be maintained from tank surface to any Air intake units, Window type and Split Type AC units, Air vents, balanced type flue outlets.
- 10 m distance shall be maintained from tanks to Fire Hydrant, firefighting equipment, firefighting valves, fire command centers.
- LPG fill line and connection arrangement shall be fully exposed outside the building. Where need arises to cover it aesthetically, fully ventilated, or fully perforated enclosure is acceptable.
- Filling connections shall be electrically classified areas without any ignition sources within 7.6 m.

FIRE DETECTION & ALARM SYSTEM

In accordance with NFPA 101 Section 30.3.4.1.1, Apartment buildings four or more stories in height or with more than 11 dwelling units shall be provided with a fire alarm system.

In accordance with Chapter 8 Table 8.13 UAE Fire and Life Safety Code, as proposed project is classified high-rise building for accommodation buildings, smoke detection and alarm system shall be provided throughout the building.

In accordance with NFPA 101 Section 30.3.4.1.1 and 9.6.1.7, a complete fire alarm system shall provide functions for initiation, notification, and control, which shall perform as follows:

- Initiation - The initiation function provides the input signal to the system.
- Notification - The notification function is the means by which the system advises that human action is required in response to a particular condition.
- Detection & Control - The control function provides outputs to control building equipment to enhance protection of life.

Initiating Devices for Fire Alarm System

The initiation of the fire alarm system shall be by any or all of the following means:

- Activation of automatic smoke and / or heat detectors
- Activation of a manual call point
- Operation of an extinguishing system
- As a result of the fire alarm system not being reset during an investigation period

Automatic Detectors

Smoke, heat and multi-sensor detectors shall be provided throughout the apartment building, service rooms, auxiliary rooms & common areas and located in accordance with NFPA 72. As a minimum these shall be installed inside apartments, corridors, lobbies, gymnasiums and assembly areas.

Pump rooms, gymnasiums involving dusts or vapours shall be protected by heat-detection devices. Refer Figure below shows the placement of smoke detectors and heat detectors in smooth ceilings.

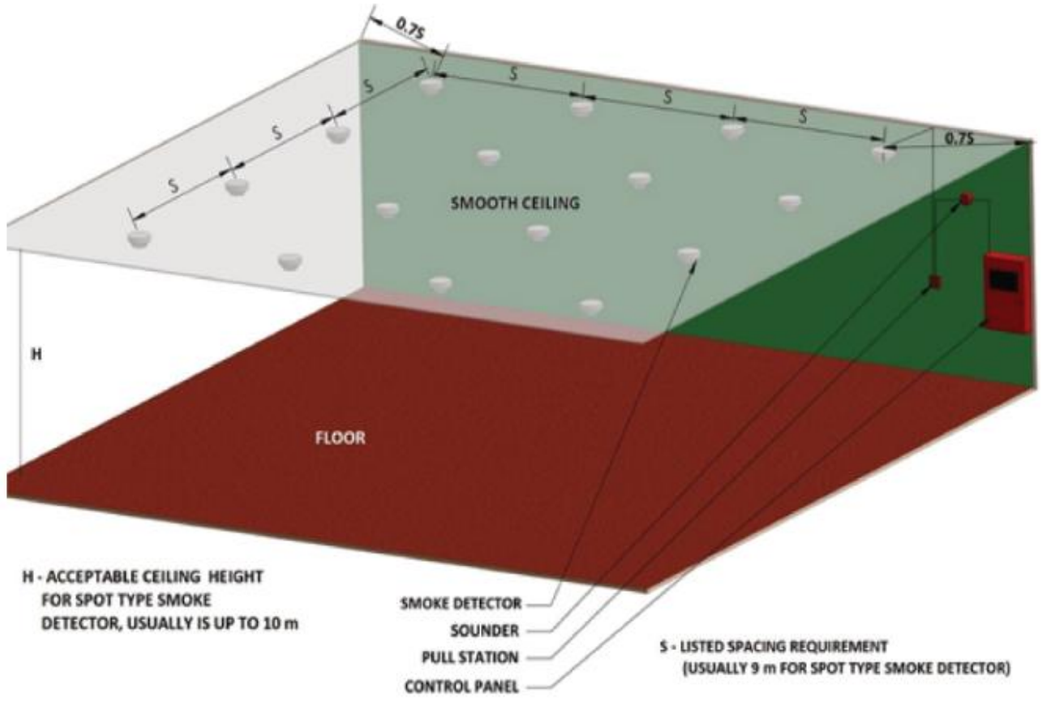


Figure 30: Smoke Detectors on Smooth Ceilings

In high-rise buildings, along with one smoke detector at the top of the stair, multiple smoke detectors at intermediate landings, every 23m, shall be provided as per UFC -2018.

In high-rise buildings, along with one smoke detector at the top of the shaft, multiple smoke detectors at intermediate levels, every 23m, shall be provided to initiate elevator recall as per UFC -2018.

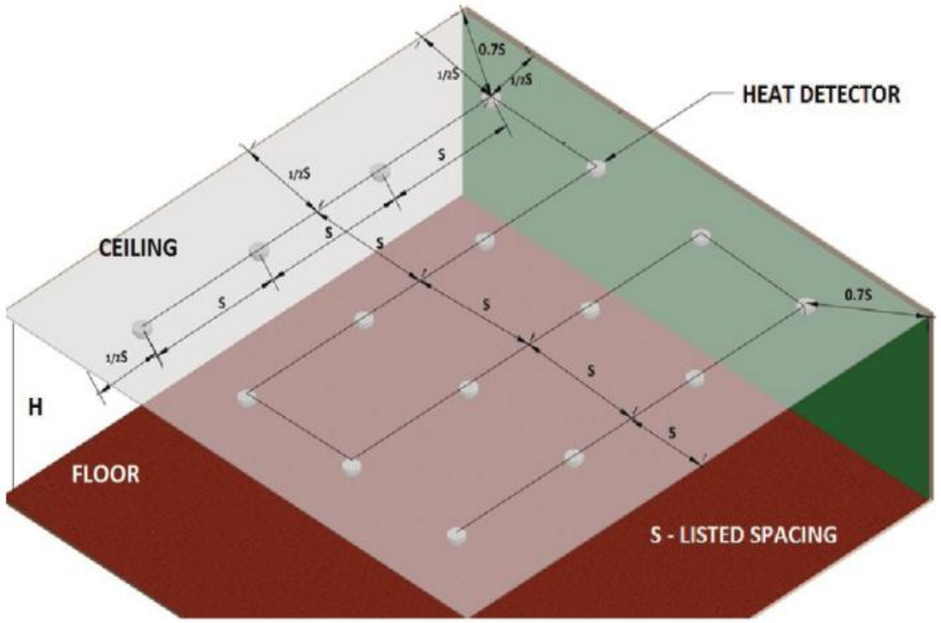


Figure 31: Heat detectors on smooth ceilings

Smoke Alarms

In accordance with NFPA 101, Section 30.3.4.5 smoke alarms shall be installed in every sleeping area, outside every sleeping area in the immediate vicinity of the bedrooms, and on all levels of the dwelling unit.

Manual Call Points

The manual call points shall be installed on all escape routes and beside stairwell entrances and all exits to open air for all buildings in the project. The manual call points shall be installed so that they are conspicuous, unobstructed, and accessible. Distribution of the manual call points should be such that travel distance should not be more than 61 m to reach the nearest manual call point.

Two-Way Communication System

As per UAE Fire Code 2018, following should apply:

- Two-way communication system shall be installed for communication between the emergency/fire command centre (ECC) and:
  - Inside stair, on every floor landing
  - Inside each elevator cars
  - Inside Fire Pump Room
  - Inside Fire Lift Lobby
  - Inside Area of Refuge, if any
- At least two handsets shall be stored at each control center for use by emergency responders.
- Each telephone station or telephone jack shall be clearly and permanently labeled to allow the caller to identify his or her location to the control center by voice.

As per NFPA 72 Sec 7.14.3.4, the following should apply to the two-way communication service:

- The system should be listed as a two-way telephone communication service.

- The system should be monitored for integrity, as per NFPA 72 requirements.
- Up to 5 telephones can be used simultaneously.
- It should operate between the emergency/fire command centre, elevator cabs, elevator lobbies, emergency and standby power rooms, fire pump room, and at entries into enclosed stairways at every floor.
- All phone jacks should have spring-loaded flip-up (weatherproof) covers.
- A speaker, with adjustable volume control, for monitoring conversations on this phone system should be provided in the emergency/fire command centre.
- A minimum of ten portable handsets should be provided in a labelled cabinet inside the panel in the emergency/fire command centre.
- The system wiring should be shielded from induction noise and be properly separated from other wiring in accordance with NFPA 70.
- Wall-mounted telephone appliances or related jacks should be not less than 910 mm and not more than 1.68 m above floor level with clear access to the appliance that is at least 760 mm wide. If accessible to the general public, one telephone appliance per location should be not more than 1.22 m above floor level.
- Audible and visual indicator is required at the panel when the phone is plugged into any phone jack.



Figure 32: Two - way Telephone Jack for Fire Fighters for illustration

Fire Alarm Control Panel

As per UAE Fire Code, Chapter 8, Table 8.13 - Main FACP shall be located at the continuously attended location such as emergency/fire command centre or reception or Security guard house, as per the building fire strategy.

All components of the fire alarm system shall be coordinated to ensure that the main fire alarm panel can receive a fire and/ or fault signal from all areas as well as from the fire suppression systems prior to initiating an evacuation signal and activating fire related plant.

As per UAE Fire Code, Chapter 8 Table 8.13 For High-rise Building - Additional sub FACP's or annunciators complete with all functional features shall be provided at the elevator lobby of the floor or the mechanical floor at every 45m height of the building and connected to main FACP in a network.

The repeater and annunciator panels, if provided shall be connected to the main Fire Alarm Control panel.

EMERGENCY LIGHTING AND EXIT SIGNAGE

Emergency Lighting

The required emergency lighting and exit signage shall be provided in accordance with the latest UAE Fire Code Chapter 6.
Emergency Lighting System shall be provided for the building. Emergency Lighting is required for egress paths and emergency response and service areas including

- Inside Fire Pump Room, EPS Room, and Large Plant Rooms
- Inside and Outside Emergency Command Center
- Inside and Outside Stairways
- Inside Lift Lobby
- Inside Fire Fighting Lift Lobby
- In Exit Corridors
- Inside Exit Access Corridors
- Inside Exit Passageways
- All Areas of enclosed Basements and Ramps
- In Service Corridors
- Outside Exit Discharge
- Reception
- Location of FACP
- Location of Fire Hose Cabinet
- Location of Fire Extinguishers
- Location of Smoke Control Panel
- LPG Tank Location
- Inside LV Room
- Inside Transformer Room

Emergency lighting shall provide sufficient illumination to:

- Indicate clearly and unambiguously the way out along pre-determined escape routes
- Indicate and illuminate fire alarm call points and fire extinguisher locations along escape routes

The emergency lighting system shall be arranged to provide the required illumination automatically in the event of any interruption of normal lighting for a minimum of 3 hours because of any of the following occurring:

- Failure of a public utility or other outside power supply.
- Opening of a circuit breaker or fuse.
- Manual act[s], including accidental opening of the central normal lighting facilities.

Initial illumination shall be not less than an average of 10.8 lux with a minimum of 1.1 lux at any point measured along the path of egress at floor level and shall not fall to an average of 0.65 lux after 1.5 hours. The maximum to minimum illumination uniformity ratio shall be designed not to exceed 40 to 1. The delay period between failure of normal lighting and energising the emergency lighting system shall not exceed 10 second.

Exit Signage

In accordance with Chapter 5, of UAE Fire Code, exits signs on the following location:

1. The entrance to every exit on every floor shall be clearly indicated by an exit sign placed over the exit door. Such signs shall be placed so that they are always clearly visible.
2. In long corridors, in open floor areas and in all situations where the location of the exits may not be readily visible, directional signs shall be provided to serve as guides from all portions of the corridors or floors.
3. Exits, other than main exterior exit doors that obviously and clearly are identifiable as exits, exits shall be marked by an approved sign that is readily visible from any direction of exit access.
4. Horizontal components of the egress path within an exit enclosure shall be marked by approved exit or directional exit signs where the continuation of the egress path is not obvious.
5. Access to exits shall be marked by approved, readily visible signs in all cases where the exit or way to reach the exit is not readily apparent to the occupants. A sign with a directional indicator showing the direction of travel shall be placed in every location where the direction of travel to reach the nearest exit is not apparent to the occupants.
6. Exit Signage shall be always illuminated.

STANDBY EMERGENCY POWER

The standby power system shall be a Type 60, Class 2, Level 1 system compliant with NFPA 110. Generators and battery systems shall be located in a room separated from the remainder of the building by fire barriers having a minimum 2-hour fire resistance rating. Wiring for power and control shall be located and properly protected to ensure a minimum 1 hour of operation in the event of a fire.
In accordance with DCD requirements the following standby power loads must be connected to the emergency generator and must be operational within 60 seconds.

- Fire service elevators
- Emergency mechanical air handling systems (i.e., smoke proof enclosures and smoke management systems)
- Electric fire pumps (where both main & standby pumps are electrical motor driven)
- Emergency command centre equipment and lighting

The following systems shall operate with no more than 10 seconds of transfer time between normal power and emergency (standby) power, batteries, or UPS:

- Exit signs and means of egress illumination.
- Elevator car lighting
- Fire detection and alarm system
- Sprinkler alarm system
- Smoke control system
- Voice communication system

Any UPS system installed in the building shall be compliant with NFPA 111.

SOLAR POWER GENERATION SYSTEMS

In accordance to chapter 14, Section 2.2.3. of UAE Fire code 2018. The Fire and Life Safety requirements for Solar Power Generation Systems shall be as per Table 14.2. following shall be complied with.

1. FLAME SPREAD CHARACTERS-

1. FLAME SPREAD CHARACTERISTICS OF PV CELLS ON ROOF

i. The minimum requirement for Solar PV cells to be installed on roof of Lowrise buildings and Private Villas is Class B or equivalent class when tested to test standards as required by Section 2.2.4. of chapter 14, UAE Fire code 2018.

ii. For Midrise, Highrise and Super Highrise buildings, the minimum requirement for solar PV cells to be installed on roof is Class A, when tested to test standards as per Section 2.2.4. of chapter 14, UAE Fire code 2018.

iii. Where solar panels are installed on combustible roofs, such combustible roofs shall be separated from the solar panels, cabling and installations by noncombustible material in accordance with Chapter 1. Section 7.1.44.

2. FLAME SPREAD CHARACTERISTICS OF PV CELLS ON FACADE

i. The minimum requirement for Solar PV cells to be installed on building façade of any building of any occupancy shall be Class A, when tested to test criteria as per Section 2.2.4. of chapter 14, UAE Fire code 2018.

ii. When solar panels, PV cells are installed on the façade systems, they shall not reduce the structural integrity or the fire resistance rating of the exterior walls. iii. The solar panels and PV cells installed on building envelope, shall demonstrate through large scale system tests that the building envelope is not altered in its fire resistance rating. See Section 2.2.4. of chapter 14, UAE Fire code 2018 for fire test requirements.

2. CONSTRUCTION-

1. ACCESS

i. Solar panel installations on roof or on commercial scale as generation plant shall provide access to fire fighters around the installations.

ii. Minimum of 1.2 m wide access path shall be available around the installations.

iii. Solar installations on roof shall be such that access and emergency operations from first responders and fire fighters operations and maneuvering is not jeopardized.

iv. Commercial installations on ground such as Solar power plants shall have 6 m clear fire access around the installations and in between arrays in full compliance with Chapter 2. Fire Access, taking into account the dead ends, maximum distance, turning radius etc.

2. SECURITY

i. Commercial Solar Panel installations shall be secured with fence and CCTV monitoring to prevent accidental entry of people and animals. Accidental trespassing of people and animals into such installations could result in loss of life and damage to installations.

ii. In domestic installations such as on roofs, facility management should take precautions to isolate and restrict access to such areas to prevent accidental trespassing of children and people, endangering their lives.

3. LOCATION

i. The solar panel installations on roof shall not obstruct the Smoke control system (Pressurization system, smoke extract system etc.) equipment and components such as air intake openings and fans. Obstructions to smoke control system components can severely affect the efficiency and intent of the smoke control strate-gy of the buildings.

ii. solar panel installations shall not obstruct any building means of egress.

iii. Solar panel installations shall not obstruct access to LPG Tanks, if any.

iv. Solar panel installations shall not obstruct building HVAC components.

4. MEANS OF EGRESS

i. Other than Private Villas and Commercial Villas, solar installations shall have access to two permanent building exit stairs, designed as per Chapter 3. Means of Egress.
ii. Solar installations having access with temporary ladders, spiral stairs and open steel stairs are not acceptable.

3. SUBSTATIONS-

i. Substations, if any, shall comply with Section 2.1. of chapter 14, UAE Fire code 2018.

4. EQUIPMENT-

i. Transformers, if any, shall comply with Section 2.1. of chapter 14, UAE Fire code 2018.
ii. Inverters shall be listed and approved as per Section 2.2.3.7.

5. SYSTEM IDENTIFICATION-

i. Equipment and the system shall have clear identification label near the control system or Command Center or reception, stating if the type of solar system is “Thermal” or “Photovoltaic” and the associated warning and safety precautions to be taken during fire accidents.
ii. Solar components release hazardous and toxic elements when they decompose in fire. These warning details shall be identified and made visible with warning signs.
iii. Color coding and standard identification of energized parts of solar system shall be in place to identify “ON” parts such as electrical wire conduits, panel segments etc.

6. EMERGENCY SHUTDOWN-

i. In large solar power generation installations, adequate module and array level isolation (disconnect facilities) shall be provided segment wise strategically. Remote shutdown features of modules and arrays shall be considered to enable emergency responders and fire fighters, to safely isolate energized segments and successfully execute fire fighting operations.
ii. In domestic applications, module and array disconnect switches shall be provided on roof.
iii. Disconnect switches for solar system shall be clearly labeled and distinct from any other devices such as “Manual Call Points”, “Emergency LPG Shut-off” and activation buttons of “Fire suppression systems”.
iv. Location of disconnect switches, inverters and arrangement of solar arrays shall be depicted in a “Site layout” plan and shall be posted at key locations such as control rooms, emergency command centers, building receptions etc.

7. FIRE DETECTION AND ALARM SYSTEM-

i. Fire Detection and Alarm shall be provided to all the ancillary buildings such as control room, equipment room, storage room and inverter rooms of the solar energy generation plants and commercial installations, in accordance with Chapter 8: Fire Detection and Alarm System.
ii. Main FACP shall be located in control room with annunciator panel at security guard room or main panel shall be located at security guard room.

8. FIRE EXTINGUISHERS-

i. In solar energy generating plants, Fire extinguishers shall be provided throughout ancillary buildings in accordance with Chapter 4: Fire Extinguishers.
ii. Trolley type, wheeled CO2 extinguishers, in accordance with Chapter 4, Table 4.3.5., shall be provided at strategic locations throughout large outdoor solar panel installations in an enclosed protected outdoor cabinet.

iii. One Trolley type, wheeled CO2 extinguisher, in accordance with Chapter 4, Table 4.3.5., shall be provided on roof, where solar panels are installed on roof.
iv. Foam Type and Water Type extinguishers shall not be used on solar installation fires.

9. FIRE PROTECTION SYSTEM-

NOT REQUIRED. However, substations and transformers, if any, shall comply with Section 2.1. of chapter 14, UAE Fire code 2018.

10. WARNING SIGNS-

i. Warning signs shall be provided near all solar system installations, as shown.



Figure 33: Solar Power Plant for illustration

11. FIRE STRATEGY-

1. GENERAL

I. Preplanned Fire strategy is of utmost important requirement in tackling solar power system fires, since it poses additional dangers and hazards to fire fighters than the normal fires.
II. The Utility service provider, stake holders, owners, system operators, manufacturers and installers shall jointly develop the fire strategy along with Civil Defence Operation Department.
III. A clear written fire strategy addressing the electrical shock hazard, isolation of panels and arrays, isolation of modules, type of fire fighting agent to be used and the method of fire fighting etc. based on the following.

2. KEY ELEMENTS TO BE CONSIDERED IN THE FIRE STRATEGY

i. Management shall assign responsible and knowledgeable person to coordinate with Emergency responders and Civil Defence, who knows hazards and disconnection switch locations and methods and who can assist Civil Defence during fire emergencies.
ii. Photovoltaic panels exposed to sun are always “ON” and hence energized.
iii. De-energizing solar panels is almost impossible. So first responders and fire fighters should always consider the solar system and its components as “energized”.
iv. Care should be taken never to cut or damage conduits and equipment and should be treated as energized always.
v. Facility management and Utility provider should always be familiar with locations of disconnection switches scattered all along the vast areas of installations.
vi. If all connections to inverter are not disconnected, the grid will still be live with voltages.
vii. Battery storage arrangement poses additional threat. Batteries maintain electrical current during night time. Batteries on fire can be explosive and emit hazardous materials, corrosive gases and toxic fumes.
viii. Appropriate PPE (Personal Protective Equipment) shall be available at all times along with respiratory protective equipment (Self-contained Breathing Apparatus), during and post fire clean up activities.

ix. Minimum of 5 m distance should be maintained from these equipment while fighting fire.

3. KEY ELEMENTS TO BE CONSIDERED IN THE POST FIRE STRATEGY

i. During normal operation, solar power components are safe but during fire and under fire solar cells, components are very dangerous in terms of hazardous and toxic materials emitted.
ii. Care should be taken to avoid exposure to these exposures during fire and during clean up activities.
iii. New solar technologies in domestic usage are integral part of the building components and materials and during fire, the hazardous nature of these materials may not be obvious to occupants, first responders and fire fighters.

12. TRAINING-

i. The solar power generation organization, service providers along with manufacturers shall regularly conduct training programs and workshops for the maintenance contractors, emergency responders and Civil Defence personnel.
ii. Training shall concentrate on equipping emergency responders and Civil Defence personnel with in-depth knowledge of Solar cells such as:

a. Their behavior under fire scenarios
b. Burning characteristics
c. Flame spread, smoke generation, hazardous, corrosive and toxic gases generated etc.
d. Identification of energized parts of the system
e. Identification of disconnecting switches and methods
f. International best practices of solar system fire fighting

Refer to chapter 14, Section 2.2.4. (Photovoltaic Cell, PV) Material Test Standards and Approval, which shall be followed and complied with.

INFORMATION USED AND LIMITATIONS OF THIS REPORT

This report is based on drawings received from KEO for Expo Hills Residences located in Expo City, Dubai -UAE.
The recommendations are based on the information made available by the design team to us. No liability is accepted on the accuracy of the information provided. Any change in the above information to suit future reorganization or planning will require further assessment to confirm compliance with the intent of the design objectives.

This report considers the requirements of the most recent UAE Fire and Life Safety Code (2018) / NFPA Life Safety Code as mentioned earlier and as such deals with fire protection and life safety for entire project. The recommendations of the report are subject to final approval by Dubai Civil Defence (DCD).

This report does not take into consideration future potential new amendments to UAE Fire Code / NFPA. However, the features that are included for life safety to gain building regulations approval will contribute to some extent to business and property protection.

This report is prepared in good faith and with due care for information purposes only and should not be relied upon as providing any warranty or guarantee that ignition or a fire will not occur.
The report does not specifically consider property damage such as building and contents damage caused by fire, potential increased insurance liability and loss of business continuity.

The information contained within this report is for use solely in relation to the proposed project and shall not be used in relation to any other project. The report is not for the use of or reliance on, by third parties.

APPENDIX A

- **Occupancy Load Calculation**

Occupant Load Table																																
Level	Room Name	Floor Area (m2)	OL factor	Occupant Load	Total OL/fl.	No. of stairs /fl.	OL stair #1 (TOWER A1)	OL stair #2 (TOWER A1)	OL stair #3 (TOWER A1)	OL stair #4 (TOWER A1)	OL stair #1 (TOWER A2)	OL stair #2 (TOWER A2)	OL stair #3 (TOWER A2)	OL stair #1 (TOWER A3)	OL stair #2 (TOWER A3)	OL stair #3 (TOWER A3)	OL stair #1 (TOWER A4)	OL stair #2 (TOWER A4)	OL stair #3 (TOWER A4)	OL stair #1 (TOWER A5)	OL stair #2 (TOWER A5)	OL stair #3 (TOWER A5)	OL stair #1 (TOWER A6)	OL stair #2 (TOWER A6)	OL stair #3 (TOWER A6)	OL stair #1 (TOWER A7)	OL stair #2 (TOWER A7)	OL stair #1 (TOWER A8)	OL stair #2 (TOWER A8)	OL stair #3 (TOWER A8)	Stair Clear Width (mm)	Total Capacity /Stair	Total Exit Stair Capacity/Floor
BASEMENT FLOOR	Car Parking and Ancialiary Services	59962	27.9	2149	2168	17	128	128	-	-	128	128	-	128	128	-	128	128	-	128	128	-	128	128	-	128	128	128	128	128	1200	158	2684
	Security Control Room	64	9.3	7																												
	FM IT / CAFM ROOM 1	19	9.3	2																												
	FM IT / CAFM ROOM 2	28	9.3	3																												
	FM IT / CAFM ROOM 3	34	9.3	4																												
	FM IT / CAFM ROOM 4	15	9.3	2																												
	Security Office	16	9.3	2																												
OVERALL GROUND FLOOR	Car Parking, Storage Rooms, MEP Rooms and Ancialiary Services	47599	27.9	1706	4409	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	North Lobby Seating - Tower A1	18	1.4	13																												
	North Reception - Tower A1	5	4.6	1																												
	South Lobby Seating - Tower A1	18	1.4	13																												
	South Reception - Tower A1	5	4.6	1																												
	FCC - Tower A1	40	9.3	4																												
	FM Office / Helpdesk - Tower A1	20	9.3	2																												
	Briefing/Induction Room - Tower A1	19	9.3	2																												
	North Lobby Seating - Tower A2	18	1.4	13																												
	North Reception - Tower A2	5	4.6	1																												
	South Lobby Seating - Tower A2	18	1.4	13																												
	South Reception - Tower A2	5	4.6	1																												
	Supermarket - Tower A2	1134	2.8	405																												
	Prayer Room - Tower A2	14	0.9	16																												
	Pantry / Break Room - Tower A2	11	9.3	1																												
	Male Lockers - Tower A2	7	1.4	5																												
	Female Lockers - Tower A2	5	1.4	4																												
	North Lobby Seating - Tower A3	18	1.4	13																												
	North Reception - Tower A3	5	4.6	1																												
	FCC - Tower A3	35	9.3	4																												
	South Lobby Seating - Tower A3	18	1.4	13																												
	South Reception - Tower A3	5	4.6	1																												
	FM Office / Helpdesk - Tower A3	17	9.3	2																												
	Briefing/Induction Room - Tower A3	10	9.3	1																												
	Prayer Room - Tower A3	26	0.9	29																												
	F & B 02 - Tower A3	251	1.4	180																												
	F & B 01 - Tower A3	291	1.4	208																												
	North Lobby Seating - Tower A4	18	1.4	13																												
	North Reception - Tower A4	5	4.6	1																												
	Prayer Room-Tower A4	11	0.9	12																												
	Pantry / Break Room - Tower A4	11	9.3	1																												
	Male Changing/ Lockers - Tower A4	15	1.4	11																												
	Female Changing/ Lockers - Tower A4	20	1.4	14																												
	Retail 01 - Tower A4	174	2.8	62																												
	Retail 02 - Tower A4	160	2.8	57																												
	Retail 03 - Tower A4	156	2.8	56																												
	Nursery - Tower A4	602	3.3	182																												
	North Lobby Seating - Tower A5	18	1.4	13																												
	North Reception - Tower A5	5	4.6	1																												
	FCC - Tower A5	38	9.3	4																												
	Security Office - Tower A5	9	9.3	1																												
	FM Office / Helpdesk - Tower A5	20	9.3	2																												
	Briefing/Induction Room - Tower A5	17	9.3	2																												
	Retail 04 - Tower A5	124	2.8	44																												
	Retail 05 - Tower A5	215	2.8	77																												
	Retail 06 - Tower A5	144	2.8	51																												
	F & B 03 - Tower A5	285	1.4	204																												
	North Lobby Seating - Tower A6	18	1.4	13																												
	North Reception - Tower A6	5	4.6	1																												
	Prayer Room - Tower A6	9	0.9	10																												
	Pantry / Break Room - Tower A6	13	9.3	1																												
	Male Changing/ Lockers - Tower A6	15	1.4	11																												
	Female Changing/ Lockers - Tower A6	20	1.4	14																												
	Retail 07 - Tower A6	142	2.8	51																												
	Retail 08 - Tower A6	184	2.8	66																												
	Retail 09 - Tower A6	118	2.8	42																												
	F & B 04 - Tower A6	272	1.4	194																												
	North Lobby Seating - Tower A7	18	1.4	13																												
	North Reception - Tower A7	5	4.6	1																												
	FCC - Tower A7	33	9.3	4																												
	Retail 10 - Tower A7	296	2.8	106																												
	F & B 05- Tower A7	295	1.4	211																												
	F & B 06- Tower A7	127	1.4	91																												
	West Lobby Seating - Tower A8	4	1.4	3																												
	West Reception - Tower A8	4	4.6	1																												
	East Lobby Seating- Tower A8	18	1.4	13																												
	East Reception - Tower A8	4	4.6	1																												
	Retail 11 - Tower A8	213	2.8	76																												
	Prayer Room - Tower A8	18	0.9	20																												
	Pantry / Break Room - Tower A8	40	9.3	4																												
	FM Office / Helpdesk - Tower A8	15	9.3	2																												
	Briefing/Induction Room - Tower A8	14	9.3	2																												

	Apartment Units (Tower 1)	3296	18.6	177																												
	Amenities Lobby Tower 1)	75	1.4	54																												
	Gym	137	4.6	30																												
	Outdoor Area 1 (Tower 1)	164	1.4	117																												
	Family and Children's Pool	368	4.6	80																												
	Pool Deck	90	2.8	32																												
	Outdoor Area 2 (Tower 1)	32	1.4	23																												
	Apartment Units (Tower 2)	2948	18.6	158																												
	Amenities Lobby (Tower 2)	76	1.4	54																												
	Outdoor Area 1 (Tower 2)	160	1.4	114																												
	Volleyball	12	no. of players	12																												
	Outdoor Area 2 (Tower 2)	91	1.4	65																												
	Seating @ Tower A2 -A3	23	0.455	51																												
	Apartment Units (Tower 3)	2681	18.6	144																												
	Amenity Lobby (Tower 3)	69	1.4	49																												
	Gym	136	4.6	30																												
	Outdoor Area 1 (Tower 3)	205	1.4	146																												
	Kids and Adult Pool	241	4.6	52																												
	Pool Deck	40	2.8	14																												
OVERALL 1ST FLOOR (TOWER A1 TO A8)	Outdoor Area 2 (Tower 3)	43	1.4	31	2895	24	121	121	121	121	121	121	121	121	121	121	121	121	121	121	121	121	121	121	121	121	121	121	121	1200	158	3789
	Outdoor Seating Area (Tower 3)	20	1.4	14																												
	Apartment Units (Tower 4)	2314	18.6	124																												
	Amenity Lobby (Tower 4)	68	1.4	49																												
	Outdoor Area 1 (Tower 4)	45	1.4	32																												
	Outdoor Area 2 (Tower 4)	66	1.4	47																												
	Apartment Units (Tower 5)	2055	18.6	110																												
	Amenities Lobby (Tower 5)	76	1.4	54																												
	Picnic Area (Tower 5)	12	1.4	9																												
	Gym (Tower 5)	238	4.6	52																												
	Beach Pool	141	4.6	31																												
	Pool Deck	57	2.8	20																												
	Kids Pool	32	4.6	7																												
	Apartment Units (Tower 6)	1690	18.6	91																												
	Amenities Lobby (Tower 6)	76	1.4	54																												
	Outdoor Area (Tower 6)	132	1.4	94																												
	Apartment Units (Tower 7)	1620	18.6	87																												
	Amenities Lobby (Tower 7)	66	1.4	47																												
	Island Pool	213	4.6	46																												
	Pool Deck	28	2.8	10																												
	Cabanas	28	1.4	20																												
	Apartment Units (Tower 8)	3157	18.6	170																												
	Amenities Lobby (Tower 8)	47	1.4	34																												
	Outdoor Area (Tower 8)	82	1.4	59																												
	Gym	242	4.6	53																												
	Seating (Total Length Tower 1-8)	67	0.455	147																												
2ND FLOOR - TYPICAL FLOOR	Apartment Units (Tower A1)	4108	18.6	221	221	3	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A2)	4105	18.6	221	221	3	-	-	-	-	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A3)	3835	18.6	206	206	3	-	-	-	-	-	-	-	69	69	69	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A4)	3482	18.6	187	187	3	-	-	-	-	-	-	-	-	-	-	62	62	62	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A5)	2872	18.6	154	154	3	-	-	-	-	-	-	-	-	-	-	-	-	-	51	51	51	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A6)	2515	18.6	135	135	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	45	45	45	-	-	-	-	1200	158	474
	Apartment Units (Tower A7)	2179	18.6	117	117	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	-	-	1200	158	316
	Apartment Units (Tower A8)	3318	18.6	178	178	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	1200	158	474
3RD FLOOR - TYPICAL FLOOR	Apartment Units (Tower A1)	4108	18.6	221	221	3	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A2)	4105	18.6	221	221	3	-	-	-	-	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A3)	3835	18.6	206	206	3	-	-	-	-	-	-	-	69	69	69	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A4)	3482	18.6	187	187	3	-	-	-	-	-	-	-	-	-	-	62	62	62	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A5)	2872	18.6	154	154	3	-	-	-	-	-	-	-	-	-	-	-	-	-	51	51	51	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A6)	2515	18.6	135	135	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	45	45	45	-	-	-	-	1200	158	474
	Apartment Units (Tower A7)	2179	18.6	117	117	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	-	-	1200	158	316
	Apartment Units (Tower A8)	3318	18.6	178	178	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	1200	158	474
4TH FLOOR - TYPICAL FLOOR	Apartment Units (Tower A1)	4108	18.6	221	221	3	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A2)	4105	18.6	221	221	3	-	-	-	-	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A3)	3835	18.6	206	206	3	-	-	-	-	-	-	-	69	69	69	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A4)	3482	18.6	187	187	3	-	-	-	-	-	-	-	-	-	-	62	62	62	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A5)	2872	18.6	154	154	3	-	-	-	-	-	-	-	-	-	-	-	-	-	51	51	51	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A6)	2515	18.6	135	135	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	45	45	45	-	-	-	-	1200	158	474
	Apartment Units (Tower A7)	2179	18.6	117	117	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	-	-	1200	158	316
	Apartment Units (Tower A8)	3318	18.6	178	178	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	1200	158	474
5TH FLOOR - TYPICAL FLOOR	Apartment Units (Tower A1)	4108	18.6	221	221	3	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A2)	4105	18.6	221	221	3	-	-	-	-	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A3)	3835	18.6	206	206	3	-	-	-	-	-	-	-	69	69	69	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A4)	3482	18.6	187	187	3	-	-	-	-	-	-	-	-	-	-	62	62	62	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A5)	2872	18.6	154	154	3	-	-	-	-	-	-	-	-	-	-	-	-	-	51	51	51	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A6)	2515	18.6	135	135	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	45	45	45	-	-	-	-	1200	158	474
	Apartment Units (Tower A7)	2179	18.6	117	117	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	-	-	1200	158	316
	Apartment Units (Tower A8)	3318	18.6	178	178	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	1200	158	474
6TH FLOOR - TYPICAL FLOOR	Apartment Units (Tower A1)	4108	18.6	221	221	3	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A2)	4105	18.6	221	221	3	-	-	-	-	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A3)	3835	18.6	206	206	3	-	-	-	-	-	-	-	69	69	69	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A4)	3482	18.6	187	187	3	-	-	-	-	-	-	-	-	-	-	62	62	62	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A5)	2872	18.6	154	154	3	-	-	-	-	-	-	-	-	-	-	-	-	-	51	51	51	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A6)	2515	18.6	135	135	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	45	45	45	-	-	-	-	1200	158	474
	Apartment Units (Tower A7)	2179	18.6	117	117	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	-	-	1200	158	316
	Apartment Units (Tower A8)	3318	18.6	178	178	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	1200	158	474

7TH FLOOR - TYPICAL FLOOR	Apartment Units (Tower A1)	4108	18.6	221	221	3	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A2)	4105	18.6	221	221	3	-	-	-	-	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A3)	3835	18.6	206	206	3	-	-	-	-	-	-	-	69	69	69	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A4)	3482	18.6	187	187	3	-	-	-	-	-	-	-	-	-	-	-	62	62	62	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A5)	2872	18.6	154	154	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	51	51	51	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A6)	2515	18.6	135	135	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	45	45	45	-	-	-	1200	158	474
	Apartment Units (Tower A7)	2179	18.6	117	117	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	-	1200	158	316
	Apartment Units (Tower A8)	3318	18.6	178	178	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	1200	158	474
8TH FLOOR - TYPICAL FLOOR	Apartment Units (Tower A1)	4108	18.6	221	221	3	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A2)	4105	18.6	221	221	3	-	-	-	-	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A3)	3835	18.6	206	206	3	-	-	-	-	-	-	-	69	69	69	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A4)	3482	18.6	187	187	3	-	-	-	-	-	-	-	-	-	-	-	62	62	62	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A5)	2872	18.6	154	154	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	51	51	51	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A6)	2515	18.6	135	135	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	45	45	45	-	-	-	1200	158	474
	Apartment Units (Tower A7)	2179	18.6	117	117	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	-	1200	158	316
	Apartment Units (Tower A8)	3318	18.6	178	178	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	1200	158	474
9TH FLOOR - TYPICAL FLOOR	Apartment Units (Tower A2)	4105	18.6	221	221	3	-	-	-	-	74	74	74	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A3)	3835	18.6	206	BY	3	-	-	-	-	-	-	-	69	69	69	-	-	-	-	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A4)	3482	18.6	187	187	3	-	-	-	-	-	-	-	-	-	-	-	62	62	62	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A5)	2872	18.6	154	154	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	51	51	51	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A6)	2515	18.6	135	135	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	45	45	45	-	-	-	1200	158	474
	Apartment Units (Tower A7)	2179	18.6	117	117	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	-	1200	158	316
	Apartment Units (Tower A8)	3318	18.6	178	178	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	1200	158	474
10TH FLOOR - TYPICAL FLOOR	Apartment Units (Tower A4)	3482	18.6	187	187	3	-	-	-	-	-	-	-	-	-	-	-	62	62	62	-	-	-	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A5)	2872	18.6	154	154	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	51	51	51	-	-	-	-	-	-	1200	158	474
	Apartment Units (Tower A6)	2515	18.6	135	135	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	45	45	45	-	-	-	1200	158	474
	Apartment Units (Tower A7)	2179	18.6	117	117	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	-	1200	158	316
	Apartment Units (Tower A8)	3318	18.6	178	178	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	1200	158	474
11TH FLOOR - TYPICAL FLOOR	Apartment Units (Tower A6)	2515	18.6	135	135	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	45	45	45	-	-	-	1200	158	474
	Apartment Units (Tower A7)	2179	18.6	117	117	2	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	59	-	1200	158	316
	Apartment Units (Tower A8)	3318	18.6	178	178	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	1200	158	474
12TH FLOOR - TYPICAL FLOOR	Apartment Units (Tower A8)	3318	18.6	178	178	3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	59	1200	158	474
OVERALL ROOF FLOOR	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Occupant Load below Ground Floor		**2168**																															
Occupant Load above Ground Floor		**15211**																															
Total Occupant Load per Staircase							**636**	**636**	**636**		**709**	**709**	**709**	**670**	**670**	**670**	**682**	**682**	**682**	**584**	**584**	**584**	**571**	**571**	**571**	**706**	**706**	**775**	**775**	**775**			

EXPO CITY DUBAI | KEO

KEO SHE